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Algebraic expressions

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1.1 Introduction

EMA2

Over human history, all peoples and cultures have contributed to the field of Mathematics. Topics like algebra may seem obvious now, but for many centuries mathematicians had to make do without it. Over the next three grades, you will explore more advanced and abstract mathematics. It may not be obvious how this applies to everyday life, but the truth is, mathematics is required for nearly everything you will do one day. Enjoy your mathematical journey. Remember, there is no such thing as a “maths person” or “not a maths person”. We can all do mathematics, it just takes practice.

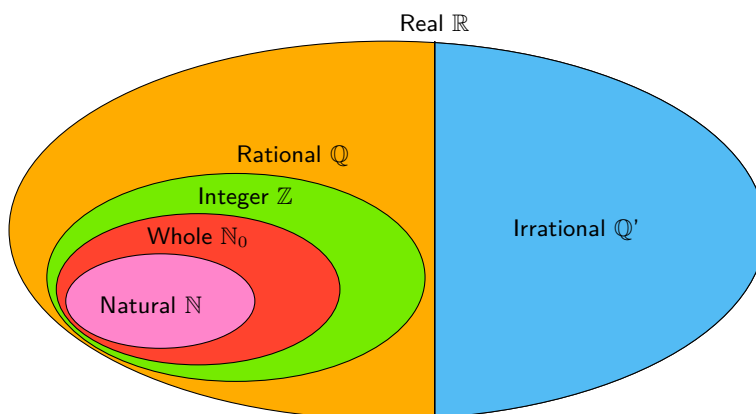


Figure 1.1: Some examples of early tally sticks. These were used to help people count things such as the number of days between events or the number of livestock they had.

In this chapter, we will begin by revising the real number system and then learn about estimating surds and rounding real numbers. We will also be expanding on prior knowledge of factorisation and delve into more complex calculations involving binomial and trinomial expressions.

1.2 The real number system

EMA3



We use the following definitions:

- $\mathbb{N}$: natural numbers are $\{1; 2; 3; \dots\}$
- $\mathbb{N}_0$: whole numbers are $\{0; 1; 2; 3; \dots\}$
- $\mathbb{Z}$: integers are $\{\dots; -3; -2; -1; 0; 1; 2; 3; \dots\}$

VISIT:

The following video shows an example of determining which of the above sets of numbers a particular number is in.

▶ See video: 2DBH at www.everythingmaths.co.za

NOTE:

Not all numbers are real numbers. The square root of a negative number is called a non-real or imaginary number. For example $\sqrt{-1}$, $\sqrt{-28}$ and $\sqrt{-5}$ are all non-real numbers.

1.3 Rational and irrational numbers

EMA4

DEFINITION: *Rational number*

A rational number ($\mathbb{Q}$) is any number which can be written as:

$$\frac{a}{b}$$

where a and b are integers and $b \neq 0$.

The following numbers are all rational numbers:

$$\frac{10}{1}; \frac{21}{7}; \frac{-1}{-3}; \frac{10}{20}; \frac{-3}{6}$$

We see that all numerators and all denominators are integers.

This means that all integers are rational numbers, because they can be written with a denominator of 1.

DEFINITION: *Irrational numbers*

Irrational numbers ($\mathbb{Q}'$) are numbers that cannot be written as a fraction with the numerator and denominator as integers.

Examples of irrational numbers:

$$\sqrt{2}; \sqrt{3}; \sqrt[3]{4}; \pi; \frac{1+\sqrt{5}}{2}$$

These are not rational numbers, because either the numerator or the denominator is not an integer.

Decimal numbers

EMA5

All integers and fractions with integer numerators and non-zero integer denominators are rational numbers. Remember that when the denominator of a fraction is zero then the fraction is undefined.

You can write any rational number as a decimal number but not all decimal numbers are rational numbers. These types of decimal numbers are rational numbers:

- Decimal numbers that end (or terminate). For example, the fraction $\frac{4}{10}$ can be written as 0,4.
- Decimal numbers that have a repeating single digit. For example, the fraction $\frac{1}{3}$ can be written as $0,\dot{3}$ or $0,\overline{3}$. The dot and bar notations are equivalent and both represent recurring 3's, i.e. $0,\dot{3} = 0,\overline{3} = 0,333\dots$
- Decimal numbers that have a recurring pattern of multiple digits. For example, the fraction $\frac{2}{11}$ can also be written as $0,\overline{18}$. The bar represents a recurring pattern of 1's and 8's, i.e. $0,\overline{18} = 0,181818\dots$

NOTE:

You may see a full stop instead of a comma used to indicate a decimal number. So the number 0,4 can also be written as 0.4

Notation: You can use a dot or a bar over the repeated digits to indicate that the decimal is a recurring decimal. If the bar covers more than one digit, then all numbers beneath the bar are recurring.

If you are asked to identify whether a number is rational or irrational, first write the number in decimal form. If the number terminates then it is rational. If it goes on forever, then look for a repeated pattern of digits. If there is no repeated pattern, then the number is irrational.

When you write irrational numbers in decimal form, you may continue writing them for many, many decimal places. However, this is not convenient and it is often necessary to round off.

NOTE:

Rounding off an irrational number makes the number a rational number that approximates the irrational number.

Worked example 1: Rational and irrational numbers

QUESTION

Which of the following are not rational numbers?

1. $\pi = 3,14159265358979323846264338327950288419716939937510\dots$
2. 1,4
3. 1,618033989...
4. 100
5. 1,7373737373...
6. $0,\overline{02}$

SOLUTION

1. Irrational, decimal does not terminate and has no repeated pattern.
2. Rational, decimal terminates.
3. Irrational, decimal does not terminate and has no repeated pattern.
4. Rational, all integers are rational.
5. Rational, decimal has repeated pattern.
6. Rational, decimal has repeated pattern.

Converting terminating decimals into rational numbers

EMA6

A decimal number has an integer part and a fractional part. For example, 10,589 has an integer part of 10 and a fractional part of 0,589 because $10 + 0,589 = 10,589$.

Each digit after the decimal point is a fraction with a denominator in increasing powers of 10.

For example:

- 0,1 is $\frac{1}{10}$
- 0,01 is $\frac{1}{100}$
- 0,001 is $\frac{1}{1000}$

This means that

$$\begin{aligned}10,589 &= 10 + \frac{5}{10} + \frac{8}{100} + \frac{9}{1000} \\&= \frac{10\,000}{1000} + \frac{500}{1000} + \frac{80}{1000} + \frac{9}{1000} \\&= \frac{10\,589}{1000}\end{aligned}$$

VISIT:

The following two videos explain how to convert decimals into rational numbers.

Part 1

▶ See video: [2DBJ](#) at www.everythingmaths.co.za

Part 2

▶ See video: [2DBK](#) at www.everythingmaths.co.za

Converting recurring decimals into rational numbers

EMA7

When the decimal is a recurring decimal, a bit more work is needed to write the fractional part of the decimal number as a fraction.

Worked example 2: Converting decimal numbers to fractions

QUESTION

Write $0,\dot{3}$ in the form $\frac{a}{b}$ (where a and b are integers).

SOLUTION

Step 1: Define an equation

$$\text{Let } x = 0,33333\dots$$

Step 2: Multiply by 10 on both sides

$$10x = 3,33333\dots$$

Step 3: Subtract the first equation from the second equation

$$9x = 3$$

Step 4: Simplify

$$x = \frac{3}{9} = \frac{1}{3}$$

Worked example 3: Converting decimal numbers to fractions

QUESTION

Write $5,4\dot{3}2$ as a rational fraction.

SOLUTION

Step 1: Define an equation

$$x = 5,432432432\dots$$

Step 2: Multiply by 1000 on both sides

$$1000x = 5432,432432432\dots$$

Step 3: Subtract the first equation from the second equation

$$999x = 5427$$

Step 4: Simplify

$$x = \frac{5427}{999} = \frac{201}{37} = 5\frac{16}{37}$$

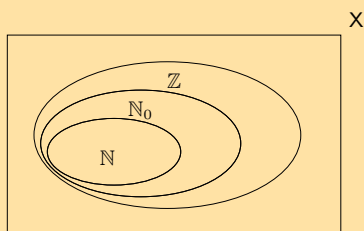
In the first example, the decimal was multiplied by 10 and in the second example, the decimal was multiplied by 1000. This is because there was only one digit recurring (i.e. 3) in the first example, while there were three digits recurring (i.e. 432) in the second example.

In general, if you have one digit recurring, then multiply by 10. If you have two digits recurring, then multiply by 100. If you have three digits recurring, then multiply by 1000 and so on.

Not all decimal numbers can be written as rational numbers. Why? Irrational decimal numbers like $\sqrt{2} = 1,4142135\dots$ cannot be written with an integer numerator and denominator, because they do not have a pattern of recurring digits and they do not terminate.

Exercise 1 – 1:

1. The figure here shows the Venn diagram for the special sets $\mathbb{N}$, $\mathbb{N}_0$ and $\mathbb{Z}$.

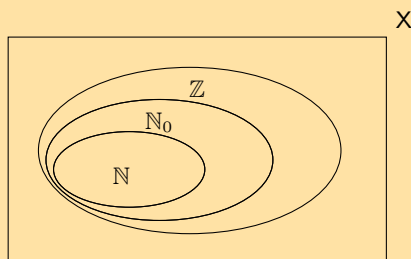


- a) Where does the number $-\frac{12}{3}$ belong in the diagram?

b) In the following list, there are two false statements and one true statement. Which of the statements is **true**?

- i. Every integer is a natural number.
- ii. Every natural number is a whole number.
- iii. There are no decimals in the whole numbers.

2. The figure here shows the Venn diagram for the special sets $\mathbb{N}$, $\mathbb{N}_0$ and $\mathbb{Z}$.



a) Where does the number $-\frac{1}{2}$ belong in the diagram?

b) In the following list, there are two false statements and one true statement. Which of the statements is **true**?

- i. Every integer is a natural number.
- ii. Every whole number is an integer.
- iii. There are no decimals in the whole numbers.

3. State whether the following numbers are real, non-real or undefined.

- a) $-\sqrt{3}$ b) $\frac{0}{\sqrt{2}}$ c) $\sqrt{-9}$ d) $\frac{-\sqrt{7}}{0}$ e) $-\sqrt{-16}$ f) $\sqrt{2}$

4. State whether the following numbers are rational or irrational. If the number is rational, state whether it is a natural number, whole number or an integer.

- a) $-\frac{1}{3}$ b) 0,651268962154862... c) $\frac{\sqrt{9}}{3}$
d) π^2 e) π^4 f) $\sqrt[3]{19}$
g) $(\sqrt[3]{1})^7$ h) $\pi + 3$ i) $\pi + 0,858408346$

5. If a is an integer, b is an integer and c is irrational, which of the following are rational numbers?

- a) $\frac{5}{6}$ b) $\frac{a}{3}$ c) $\frac{-2}{b}$ d) $\frac{1}{c}$

6. For each of the following values of a state whether $\frac{a}{14}$ is rational or irrational.

- a) 1 b) -10 c) $\sqrt{2}$ d) 2,1

7. Consider the following list of numbers:

-3 ; 0 ; $\sqrt{-1}$; $-8\frac{4}{5}$; $-\sqrt{8}$; $\frac{22}{7}$; $\frac{14}{0}$; 7 ; $1,\overline{34}$; $3,3231089\dots$; $3 + \sqrt{2}$; $9\frac{7}{10}$; π ; 11

Which of the numbers are:

- a) natural numbers
- b) irrational numbers
- c) non-real numbers
- d) rational numbers
- e) integers
- f) undefined

8. For each of the following numbers:

- write the next three digits and
- state whether the number is rational or irrational.

a) $1,1\dot{5}$

b) $2,121314\dots$

c) $1,242244246\dots$

d) $3,324354\dots$

e) $3,3243\dot{5}4$

9. Write the following as fractions:

a) $0,1$

b) $0,12$

c) $0,58$

d) $0,2589$

10. Write the following using the recurring decimal notation:

a) $0,1111111\dots$ b) $0,12121212\dots$ c) $0,123123123123\dots$ d) $0,11414541454145\dots$

11. Write the following in decimal form, using the recurring decimal notation:

a) $\frac{25}{45}$

b) $\frac{10}{18}$

c) $\frac{7}{33}$

d) $\frac{2}{3}$

e) $1\frac{3}{11}$

f) $4\frac{5}{6}$

g) $2\frac{1}{9}$

12. Write the following decimals in fractional form:

a) $0,\dot{5}$

b) $0,6\dot{3}$

c) $0,\dot{4}$

d) $5,\overline{31}$

e) $4,\overline{93}$

f) $3,\overline{93}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1. 2DBM | 2. 2DBN | 3a. 2DBP | 3b. 2DBQ | 3c. 2DBR | 3d. 2DBS | 3e. 2DBT |
| 3f. 2DBV | 4a. 2DBX | 4b. 2DBY | 4c. 2DC2 | 4d. 2DC3 | 4e. 2DC4 | 4f. 2DC5 |
| 4g. 2DC6 | 4h. 2DBZ | 4i. 2DBW | 5. 2DC7 | 6. 2DC8 | 7. 2DC9 | 8a. 2DCB |
| 8b. 2DCC | 8c. 2DCD | 8d. 2DCF | 8e. 2DCG | 9a. 2DCH | 9b. 2DCJ | 9c. 2DCK |
| 9d. 2DCM | 10a. 2DCN | 10b. 2DCP | 10c. 2DCQ | 10d. 2DCR | 11a. 2DCS | 11b. 2DCT |
| 11c. 2DCV | 11d. 2DCW | 11e. 2DCX | 11f. 2DCY | 11g. 2DCZ | 12a. 2DD2 | 12b. 2DD3 |
| 12c. 2DD4 | 12d. 2DD5 | 12e. 2DD6 | 12f. 2DD7 | | | |



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1.4 Rounding off

EMA8

Rounding off a decimal number to a given number of decimal places is the quickest way to approximate a number. For example, if you wanted to round off 2,6525272 to three decimal places, you would:

- count three places after the decimal and place a | between the third and fourth numbers;
- round up the third digit if the fourth digit is greater than or equal to 5;
- leave the third digit unchanged if the fourth digit is less than 5;
- if the third digit is 9 and needs to be rounded up, then the 9 becomes a 0 and the second digit is rounded up.

So, since the first digit after the | is a 5, we must round up the digit in the third decimal place to a 3 and the final answer of 2,6525272 rounded to three decimal places is 2,653.

VISIT:

The following video explains how to round off.

▶ See video: [2DD8](#) at www.everythingmaths.co.za

QUESTION

Round off the following numbers to the indicated number of decimal places:

1. $\frac{120}{99} = 1,1\dot{2}$ to 3 decimal places.
2. $\pi = 3,141592653\dots$ to 4 decimal places.
3. $\sqrt{3} = 1,7320508\dots$ to 4 decimal places.
4. 2,78974526 to 3 decimal places.

SOLUTION

Step 1: Mark off the required number of decimal places

If the number is not a decimal you first need to write the number as a decimal.

1. $\frac{120}{99} = 1,212|121212\dots$
2. $\pi = 3,1415|92653\dots$
3. $\sqrt{3} = 1,7320|508\dots$
4. 2,789|74526

Step 2: Check the next digit to see if you must round up or round down

1. The last digit of $\frac{120}{99} = 1,212|12121\dot{2}$ must be rounded down.
2. The last digit of $\pi = 3,1415|92653\dots$ must be rounded up.
3. The last digit of $\sqrt{3} = 1,7320|508\dots$ must be rounded up.
4. The last digit of 2,789|74526 must be rounded up.
Since this is a 9 we replace it with a 0 and round up the second last digit.

Step 3: Write the final answer

1. $\frac{120}{99} = 1,212$ rounded to 3 decimal places.
2. $\pi = 3,1416$ rounded to 4 decimal places.
3. $\sqrt{3} = 1,7321$ rounded to 4 decimal places.
4. 2,790

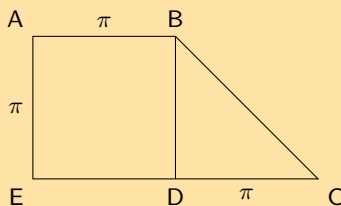
Exercise 1 – 2:

1. Round off the following to 3 decimal places:

a) 12,56637061...	b) 3,31662479...	c) 0,2666666...
d) 1,912931183...	e) 6,32455532...	f) 0,05555555...
2. Round off each of the following to the indicated number of decimal places:
 - a) 345,04399906 to 4 decimal places.
 - b) 1361,72980445 to 2 decimal places.
 - c) 728,00905239 to 6 decimal places.

- d) $\frac{1}{27}$ to 4 decimal places.
 e) $\frac{45}{99}$ to 5 decimal places.
 f) $\frac{1}{12}$ to 2 decimal places.

3. Study the diagram below



- a) Calculate the area of $ABDE$ to 2 decimal places.
 b) Calculate the area of BCD to 2 decimal places.
 c) Using your answers in (a) and (b) calculate the area of $ABCDE$.
 d) Without rounding off, what is the area of $ABCDE$?
4. Given $i = \frac{r}{600}$; $r = 7,4$; $n = 96$; $P = 200\,000$.
 a) Calculate i correct to 2 decimal places.
 b) Using your answer from (a), calculate A in $A = P(1 + i)^n$.
 c) Calculate A without rounding off your answer in (a), compare this answer with your answer in (b).
5. If it takes 1 person to carry 3 boxes, how many people are needed to carry 31 boxes?
6. If 7 tickets cost R 35,20, how much does one ticket cost?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- 1a. [2DD9](#) 1b. [2DDb](#) 1c. [2DDC](#) 1d. [2DDD](#) 1e. [2DDF](#) 1f. [2DDG](#) 2a. [2DDH](#) 2b. [2DDJ](#)
 2c. [2DDK](#) 2d. [2DDM](#) 2e. [2DDN](#) 2f. [2DDP](#) 3. [2DDQ](#) 4. [2DDR](#) 5. [2DDS](#) 6. [2DDT](#)



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1.5 Estimating surds

EMA9

If the n^{th} root of a number cannot be simplified to a rational number, we call it a surd. For example, $\sqrt{2}$ and $\sqrt[3]{6}$ are surds, but $\sqrt{4}$ is not a surd because it can be simplified to the rational number 2.

In this chapter we will look at surds of the form $\sqrt[n]{a}$ where a is any positive number, for example, $\sqrt{7}$ or $\sqrt[3]{5}$. It is very common for n to be 2, so we usually do not write $\sqrt[n]{a}$. Instead we write the surd as just $\sqrt{a}$.

It is sometimes useful to know the approximate value of a surd without having to use a calculator. For example, we want to be able to estimate where a surd like $\sqrt{3}$ is on the number line. From a calculator we know that $\sqrt{3}$ is equal to 1,73205.... It is easy to see that $\sqrt{3}$ is above 1 and below 2. But to see this for other surds like $\sqrt{18}$, without using a calculator you must first understand the following:

If a and b are positive whole numbers, and $a < b$, then $\sqrt[n]{a} < \sqrt[n]{b}$

A perfect square is the number obtained when an integer is squared. For example, 9 is a perfect square since $3^2 = 9$.

Similarly, a perfect cube is a number which is the cube of an integer. For example, 27 is a perfect cube, because $3^3 = 27$.

Consider the surd $\sqrt[3]{52}$. It lies somewhere between 3 and 4, because $\sqrt[3]{27} = 3$ and $\sqrt[3]{64} = 4$ and 52 is between 27 and 64.

VISIT:

The following video explains how to estimate a surd.

▶ See video: [2DDV](https://www.youtube.com/watch?v=2DDV) at www.everythingmaths.co.za

Worked example 5: Estimating surds

QUESTION

Find two consecutive integers such that $\sqrt{26}$ lies between them. (Remember that consecutive integers are two integers that follow one another on the number line, for example, 5 and 6 or 8 and 9.)

SOLUTION

Step 1: Use perfect squares to estimate the lower integer

$5^2 = 25$. Therefore $5 < \sqrt{26}$.

Step 2: Use perfect squares to estimate the upper integer

$6^2 = 36$. Therefore $\sqrt{26} < 6$.

Step 3: Write the final answer

$5 < \sqrt{26} < 6$

Worked example 6: Estimating surds

QUESTION

Find two consecutive integers such that $\sqrt[3]{49}$ lies between them.

SOLUTION

Step 1: Use perfect cubes to estimate the lower integer

$3^3 = 27$, therefore $3 < \sqrt[3]{49}$.

Step 2: Use perfect cubes to estimate the upper integer

$4^3 = 64$, therefore $\sqrt[3]{49} < 4$.

Step 3: Write the answer

$3 < \sqrt[3]{49} < 4$

Step 4: Check the answer by cubing all terms in the inequality and then simplify

$27 < 49 < 64$. This is true, so $\sqrt[3]{49}$ lies between 3 and 4.

Exercise 1 – 3:

1. Determine between which two consecutive integers the following numbers lie, without using a calculator:

- a) $\sqrt{18}$ b) $\sqrt{29}$ c) $\sqrt[3]{5}$ d) $\sqrt[3]{79}$ e) $\sqrt{155}$
 f) $\sqrt{57}$ g) $\sqrt{71}$ h) $\sqrt[3]{123}$ i) $\sqrt[3]{90}$ j) $\sqrt[3]{81}$

2. Estimate the following surds to the nearest 1 decimal place, without using a calculator.

- a) $\sqrt{10}$ b) $\sqrt{82}$ c) $\sqrt{15}$ d) $\sqrt{90}$

3. Consider the following list of numbers:

$$\frac{27}{7}; \sqrt{19}; 2\pi; 0,45; 0,4\overline{5}; -\sqrt{\frac{9}{4}}; 6; -\sqrt{8}; \sqrt{51}$$

Without using a calculator, rank all the numbers in ascending order.

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- 1a. [2DDW](#) 1b. [2DDX](#) 1c. [2DDY](#) 1d. [2DDZ](#) 1e. [2DF2](#) 1f. [2DF3](#) 1g. [2DF4](#) 1h. [2DF5](#)
 1i. [2DF6](#) 1j. [2DF7](#) 2a. [2DF8](#) 2b. [2DF9](#) 2c. [2DFB](#) 2d. [2DFC](#) 3. [2DFD](#)



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1.6 Products

EMAB

Mathematical expressions are just like sentences and their parts have special names. You should be familiar with the following words used to describe the parts of mathematical expressions.

$$3x^2 + 7xy - 5^3$$

Name	Examples
term	$3x^2$; $7xy$; -5^3
expression	$3x^2 + 7xy - 5^3$
coefficient	3; 7
exponent	2; 1; 3
base	x ; y ; 5
constant	3; 7; 5
variable	x ; y
equation	$3x^2 + 7xy - 5^3 = 0$

Multiplying a monomial and a binomial

EMAC

A monomial is an expression with one term, for example, $3x$ or y^2 . A binomial is an expression with two terms, for example, $ax + b$ or $cx + d$.

Worked example 7: Simplifying brackets

QUESTION

Simplify:

$$2a(a - 1) - 3(a^2 - 1)$$

SOLUTION

$$\begin{aligned}2a(a-1) - 3(a^2-1) &= 2a(a) + 2a(-1) + (-3)(a^2) + (-3)(-1) \\&= 2a^2 - 2a - 3a^2 + 3 \\&= -a^2 - 2a + 3\end{aligned}$$

Multiplying two binomials

EMAD

Here we multiply (or expand) two linear binomials:

$$(ax + b)(cx + d)$$

$$\begin{aligned}(ax + b)(cx + d) &= (ax)(cx) + (ax)d + b(cx) + bd \\&= acx^2 + adx + bcx + bd \\&= acx^2 + x(ad + bc) + bd\end{aligned}$$

Worked example 8: Multiplying two binomials

QUESTION

Find the product: $(3x - 2)(5x + 8)$

SOLUTION

$$\begin{aligned}(3x - 2)(5x + 8) &= (3x)(5x) + (3x)(8) + (-2)(5x) + (-2)(8) \\&= 15x^2 + 24x - 10x - 16 \\&= 15x^2 + 14x - 16\end{aligned}$$

The product of two identical binomials is known as the square of the binomial and is written as:

$$(ax + b)^2 = a^2x^2 + 2abx + b^2$$

If the two terms are of the form $ax + b$ and $ax - b$ then their product is:

$$(ax + b)(ax - b) = a^2x^2 - b^2$$

This product yields the difference of two squares.

Multiplying a binomial and a trinomial

EMAF

A trinomial is an expression with three terms, for example, $ax^2 + bx + c$. Now we can learn how to multiply a binomial and a trinomial.

To find the product of a binomial and a trinomial, multiply out the brackets:

$$(A + B)(C + D + E) = A(C + D + E) + B(C + D + E)$$

VISIT:

This video shows some examples of multiplying a binomial and a trinomial.

▶ See video: 2DFF at www.everythingmaths.co.za

Worked example 9: Multiplying a binomial and a trinomial

QUESTION

Find the product: $(x - 1)(x^2 - 2x + 1)$

SOLUTION

Step 1: Expand the bracket

$$(x - 1)(x^2 - 2x + 1) = x(x^2 - 2x + 1) - 1(x^2 - 2x + 1) = x^3 - 2x^2 + x - x^2 + 2x - 1$$

Step 2: Simplify

$$(x - 1)(x^2 - 2x + 1) = x^3 - 3x^2 + 3x - 1$$

Exercise 1 – 4:

1. Expand the following products:

a) $2y(y + 4)$

b) $(y + 5)(y + 2)$

c) $(2 - t)(1 - 2t)$

d) $(x - 4)(x + 4)$

e) $-(4 - x)(x + 4)$

f) $-(a + b)(b - a)$

g) $(2p + 9)(3p + 1)$

h) $(3k - 2)(k + 6)$

i) $(s + 6)^2$

j) $-(7 - x)(7 + x)$

k) $(3x - 1)(3x + 1)$

l) $(7k + 2)(3 - 2k)$

m) $(1 - 4x)^2$

n) $(-3 - y)(5 - y)$

o) $(8 - x)(8 + x)$

p) $(9 + x)^2$

q) $(-7y + 11)(-12y + 3)$

r) $(g - 5)^2$

s) $(d + 9)^2$

t) $(6d + 7)(6d - 7)$

u) $(5z + 1)(5z - 1)$

v) $(1 - 3h)(1 + 3h)$

w) $(2p + 3)(2p + 2)$

x) $(8a + 4)(a + 7)$

y) $(5r + 4)(2r + 4)$

z) $(w + 1)(w - 1)$

2. Expand the following products:

a) $(g + 11)(g - 11)$

b) $(4b - 2)(2b - 4)$

c) $(4b - 3)(2b - 1)$

d) $(6x - 4)(3x + 6)$

e) $(3w - 2)(2w + 7)$

f) $(2t - 3)^2$

- g) $(5p - 8)^2$ h) $(4y + 5)^2$ i) $(2y^6 + 3y^5)(-5y - 12)$
j) $9(8y^2 - 2y + 3)$ k) $(-2y^2 - 4y + 11)(5y - 12)$ l) $(7y^2 - 6y - 8)(-2y + 2)$
m) $(10y + 3)(-2y^2 - 11y + 2)$ n) $(-12y - 3)(2y^2 - 11y + 3)$ o) $(-10)(2y^2 + 8y + 3)$
p) $(7y + 3)(7y^2 + 3y + 10)$ q) $(a + 2b)(a^2 + b^2 + 2ab)$ r) $(x + y)(x^2 - xy + y^2)$
s) $3m(9m^2 + 2) + 5m^2(5m + 6)$ t) $4x^2(10x^3 + 4) + 4x^3(2x^2 + 6)$ u) $3k^3(k^2 + 3) + 2k^2(6k^3 + 7)$
v) $(3x + 2)(3x - 2)(9x^2 - 4)$ w) $(-6y^4 + 11y^2 + 3y)(y + 4)(y - 4)$
x) $(x + 2)(x - 3)(x^2 + 2x - 3)$ y) $(a + 2)^2 - (2a - 4)^2$

3. Expand the following products:

- a) $(2x + 3)^2 - (x - 2)^2$ b) $(2a^2 - a - 1)(a^2 + 3a + 2)$
c) $(y^2 + 4y - 1)(1 - 4y - y^2)$ d) $2(x - 2y)(x^2 + xy + y^2)$
e) $3(a - 3b)(a^2 + 3ab - b^2)$ f) $(2a - b)(2a + b)(2a^2 - 3ab + b^2)$
g) $2(3x + y)(3x - y) - (3x - y)^2$ h) $(x + y)(x - 3y) + (2x - y)^2$
i) $\left(\frac{x}{3} - \frac{3}{x}\right)\left(\frac{x}{4} + \frac{4}{x}\right)$ j) $\left(x - \frac{2}{x}\right)\left(\frac{x}{3} + \frac{4}{x}\right)$
k) $\frac{1}{2}(10x - 12y) + \frac{1}{3}(15x - 18y)$ l) $\frac{1}{2}a(4a + 6b) + \frac{1}{4}(8a + 12b)$

4. What is the value of b , in $(x + b)(x - 1) = x^2 + 3x - 4$

5. What is the value of g , in $(x - 2)(x + g) = x^2 - 6x + 8$

6. In $(x - 4)(x + k) = x^2 + bx + c$:

a) For which of these values of k will b be positive?

$-3; -1; 0; 3; 5$

b) For which of these values of k will c be positive?

$-3; -1; 0; 3; 5$

c) For what real values of k will c be positive?

d) For what real values of k will b be positive?

7. Answer the following:

a) Expand $\left(x + \frac{4}{x}\right)^2$.

b) Given that $\left(x + \frac{4}{x}\right)^2 = 14$, determine the value of $x^2 + \frac{16}{x^2}$ without solving for x .

8. Answer the following:

a) Expand: $\left(a + \frac{1}{a}\right)^2$

b) Given that $\left(a + \frac{1}{a}\right) = 3$, determine the value of $\left(a + \frac{1}{a}\right)^2$ without solving for a .

c) Given that $\left(a - \frac{1}{a}\right) = 3$, determine the value of $\left(a + \frac{1}{a}\right)^2$ without solving for a .

9. Answer the following:

a) Expand: $\left(3y + \frac{1}{2y}\right)^2$

b) Given that $3y + \frac{1}{2y} = 4$, determine the value of $\left(3y + \frac{1}{2y}\right)^2$ without solving for y .

10. Answer the following:

a) Expand: $\left(a + \frac{1}{3a}\right)^2$

b) Expand: $\left(a + \frac{1}{3a}\right)\left(a^2 - \frac{1}{3} + \frac{1}{9a^2}\right)$

c) Given that $a + \frac{1}{3a} = 2$, determine the value of $a^3 + \frac{1}{27a^3}$ without solving for a .

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2DFG	1b. 2DFH	1c. 2DFJ	1d. 2DFK	1e. 2DFM	1f. 2DFN
1g. 2DFP	1h. 2DFQ	1i. 2DFR	1j. 2DFS	1k. 2DFT	1l. 2DFV
1m. 2DFW	1n. 2DFX	1o. 2DFY	1p. 2DFZ	1q. 2DG3	1r. 2DG4
1s. 2DG5	1t. 2DG6	1u. 2DG7	1v. 2DG8	1w. 2DG9	1x. 2DGB
1y. 2DGC	1z. 2DGD	2a. 2DGF	2b. 2DGH	2c. 2DGJ	2d. 2DGK
2e. 2DGM	2f. 2DGN	2g. 2DGP	2h. 2DGQ	2i. 2DG2	2j. 2DGR
2k. 2DGS	2l. 2DGT	2m. 2DGV	2n. 2DGW	2o. 2DGX	2p. 2DGY
2q. 2DGG	2r. 2DH2	2s. 2DH3	2t. 2DH4	2u. 2DH5	2v. 2DH6
2w. 2DH7	2x. 2DH8	2y. 2DH9	3a. 2DHB	3b. 2DHC	3c. 2DHD
3d. 2DHF	3e. 2DHG	3f. 2DHH	3g. 2DHJ	3h. 2DHK	3i. 2DHM
3j. 2DHN	3k. 2DHP	3l. 2DHQ	4. 2DHR	5. 2DHS	6. 2DHT
7. 2DHV	8. 2DHW	9. 2DHX	10. 2DHY		



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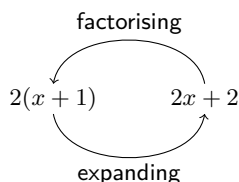


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1.7 Factorisation

EMAG

Factorisation is the opposite process of expanding brackets. For example, expanding brackets would require $2(x + 1)$ to be written as $2x + 2$. Factorisation would be to start with $2x + 2$ and end up with $2(x + 1)$.



The two expressions $2(x + 1)$ and $2x + 2$ are equivalent; they have the same value for all values of x .

In previous grades, we factorised by taking out a common factor and using difference of squares.

Factorising based on common factors relies on there being factors common to all the terms.

For example, $2x - 6x^2$ can be factorised as follows:

$$2x - 6x^2 = 2x(1 - 3x)$$

And $2(x - 1) - a(x - 1)$ can be factorised as follows:

$$(x - 1)(2 - a)$$

VISIT:

The following video shows an example of factorising by taking out a common factor.

▶ See video: [2DHZ](#) at www.everythingmaths.co.za

Worked example 10: Factorising using a switch around in brackets

QUESTION

Factorise:

$$5(a - 2) - b(2 - a)$$

SOLUTION

Use a “switch around” strategy to find the common factor.

Notice that $2 - a = -(a - 2)$

$$\begin{aligned} 5(a - 2) - b(2 - a) &= 5(a - 2) - [-b(a - 2)] \\ &= 5(a - 2) + b(a - 2) \\ &= (a - 2)(5 + b) \end{aligned}$$

Exercise 1 – 5:

Factorise:

- | | | |
|-------------------------|-------------------------|---------------------------|
| 1. $12x + 32y$ | 2. $-2ab^2 - 4a^2b$ | 3. $18ab - 3bc$ |
| 4. $12kj + 18kq$ | 5. $-12a + 24a^3$ | 6. $-2ab - 8a$ |
| 7. $24kj - 16k^2j$ | 8. $-a^2b - b^2a$ | 9. $72b^2q - 18b^3q^2$ |
| 10. $125x^6 - 5y^2$ | 11. $6x^2 + 2x + 10x^3$ | 12. $2xy^2 + xy^2z + 3xy$ |
| 13. $12k^2j + 24k^2j^2$ | 14. $3a^2 + 6a - 18$ | 15. $7a + 4$ |

For more exercises, visit www.everythingmaths.co.za and click on ‘Practise Maths’.

- | | | | | | | | |
|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------|
| 1. 2DJ2 | 2. 2DJ3 | 3. 2DJ4 | 4. 2DJ5 | 5. 2DJ6 | 6. 2DJ7 | 7. 2DJ8 | 8. 2DJ9 |
| 9. 2DJB | 10. 2DJC | 11. 2DJD | 12. 2DJF | 13. 2DJG | 14. 2DJH | 15. 2DJJ | |



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We have seen that $(ax + b)(ax - b)$ can be expanded to $a^2x^2 - b^2$.

Therefore $a^2x^2 - b^2$ can be factorised as $(ax + b)(ax - b)$.

For example, $x^2 - 16$ can be written as $x^2 - 4^2$ which is a difference of two squares. Therefore, the factors of $x^2 - 16$ are $(x - 4)$ and $(x + 4)$.

To spot a difference of two squares, look for expressions:

- consisting of two terms;
- with terms that have different signs (one positive, one negative);
- with each term a perfect square.

For example: $a^2 - 1$; $4x^2 - y^2$; $-49 + p^4$.

VISIT:

The following video explains factorising the difference of two squares.

▶ See video: [2DJK](https://www.youtube.com/watch?v=2DJK) at www.everythingmaths.co.za

Worked example 11: The difference of two squares

QUESTION

Factorise: $3a(a^2 - 4) - 7(a^2 - 4)$.

SOLUTION

Step 1: Take out the common factor $(a^2 - 4)$

$$3a(a^2 - 4) - 7(a^2 - 4) = (a^2 - 4)(3a - 7)$$

Step 2: Factorise the difference of two squares $(a^2 - 4)$

$$(a^2 - 4)(3a - 7) = (a - 2)(a + 2)(3a - 7)$$

Exercise 1 – 6:

Factorise:

1. $4(y - 3) + k(3 - y)$

2. $a^2(a - 1) - 25(a - 1)$

3. $bm(b + 4) - 6m(b + 4)$

4. $a^2(a + 7) + 9(a + 7)$

5. $3b(b - 4) - 7(4 - b)$

6. $3g(z + 6) + 2(6 + z)$

7. $4b(y + 2) + 5(2 + y)$

8. $3d(r + 5) + 14(5 + r)$

9. $(6x + y)^2 - 9$

10. $4x^2 - (4x - 3y)^2$

11. $16a^2 - (3b + 4c)^2$

12. $(b - 4)^2 - 9(b - 5)^2$

13. $4(a - 3)^2 - 49(4a - 5)$

14. $16k^2 - 4$

15. $a^2b^2c^2 - 1$

16. $\frac{1}{9}a^2 - 4b^2$

17. $\frac{1}{2}x^2 - 2$

18. $y^2 - 8$

19. $y^2 - 13$

20. $a^2(a - 2ab - 15b^2) - 9b^2(a^2 - 2ab - 15b^2)$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. [2DJM](#)2. [2DJN](#)3. [2DJP](#)4. [2DJQ](#)5. [2DJR](#)6. [2DJS](#)7. [2DJT](#)8. [2DJV](#)9. [2DJW](#)10. [2DJX](#)11. [2DJY](#)12. [2DJZ](#)13. [2DK2](#)14. [2DK3](#)15. [2DK4](#)16. [2DK5](#)17. [2DK6](#)18. [2DK7](#)19. [2DK8](#)20. [2DK9](#)
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Factorising by grouping in pairs

EMAK

The taking out of common factors is the starting point in all factorisation problems. We know that the factors of $3x + 3$ are 3 and $(x + 1)$. Similarly, the factors of $2x^2 + 2x$ are $2x$ and $(x + 1)$. Therefore, if we have an expression:

$$2x^2 + 2x + 3x + 3$$

there is no common factor to all four terms, but we can factorise as follows:

$$(2x^2 + 2x) + (3x + 3) = 2x(x + 1) + 3(x + 1)$$

We can see that there is another common factor $(x + 1)$. Therefore, we can write:

$$(x + 1)(2x + 3)$$

We get this by taking out the $(x + 1)$ and seeing what is left over. We have $2x$ from the first group and $+3$ from the second group. This is called factorising by grouping.

Worked example 12: Factorising by grouping in pairs

QUESTION

Find the factors of $7x + 14y + bx + 2by$.

SOLUTION

Step 1: There are no factors common to all terms

Step 2: Group terms with common factors together

7 is a common factor of the first two terms and b is a common factor of the second two terms. We see that the ratio of the coefficients $7 : 14$ is the same as $b : 2b$.

$$\begin{aligned} 7x + 14y + bx + 2by &= (7x + 14y) + (bx + 2by) \\ &= 7(x + 2y) + b(x + 2y) \end{aligned}$$

Step 3: Take out the common factor $(x + 2y)$

$$7(x + 2y) + b(x + 2y) = (x + 2y)(7 + b)$$

OR

Step 4: Group terms with common factors together

x is a common factor of the first and third terms and $2y$ is a common factor of the second and fourth terms ($7 : b = 14 : 2b$).

Step 5: Rearrange the equation with grouped terms together

$$\begin{aligned} 7x + 14y + bx + 2by &= (7x + bx) + (14y + 2by) \\ &= x(7 + b) + 2y(7 + b) \end{aligned}$$

Step 6: Take out the common factor $(7 + b)$

$$x(7 + b) + 2y(7 + b) = (7 + b)(x + 2y)$$

Step 7: Write the final answer

The factors of $7x + 14y + bx + 2by$ are $(7 + b)$ and $(x + 2y)$.

Exercise 1 – 7:

Factorise the following:

- | | | |
|-----------------------------------|-------------------------------------|--------------------------------|
| 1. $6d - 9r + 2t^5d - 3t^5r$ | 2. $9z - 18m + b^3z - 2b^3m$ | 3. $35z - 10y + 7c^5z - 2c^5y$ |
| 4. $6x + a + 2ax + 3$ | 5. $x^2 - 6x + 5x - 30$ | 6. $5x + 10y - ax - 2ay$ |
| 7. $a^2 - 2a - ax + 2x$ | 8. $5xy - 3y + 10x - 6$ | 9. $ab - a^2 - a + b$ |
| 10. $14m - 4n + 7jm - 2jn$ | 11. $28r - 20x + 7gr - 5gx$ | 12. $25d - 15m + 5yd - 3ym$ |
| 13. $45q - 18z + 5cq - 2cz$ | 14. $6j - 15v + 2yj - 5yv$ | 15. $16a - 40k + 2za - 5zk$ |
| 16. $ax - bx + ay - by + 2a - 2b$ | 17. $3ax + bx - 3ay - by - 9a - 3b$ | |

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

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|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2DKB | 2. 2DKC | 3. 2DKD | 4. 2DKF | 5. 2DKG | 6. 2DKH | 7. 2DKJ |
| 8. 2DKK | 9. 2DKM | 10. 2DKN | 11. 2DKP | 12. 2DKQ | 13. 2DKR | 14. 2DKS |
| 15. 2DKT | 16. 2DKV | 17. 2DKW | | | | |



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Factorising is the reverse of calculating the product of factors. In order to factorise a quadratic, we need to find the factors which, when multiplied together, equal the original quadratic.

Consider a quadratic expression of the form $ax^2 + bx$. We see here that x is a common factor in both terms. Therefore $ax^2 + bx$ factorises as $x(ax + b)$. For example, $8y^2 + 4y$ factorises as $4y(2y + 1)$.

Another type of quadratic is made up of the difference of squares. We know that:

$$(a + b)(a - b) = a^2 - b^2$$

So $a^2 - b^2$ can be written in factorised form as $(a + b)(a - b)$.

This means that if we ever come across a quadratic that is made up of a difference of squares, we can immediately write down the factors. These types of quadratics are very simple to factorise. However, many quadratics do not fall into these categories and we need a more general method to factorise quadratics.

We can learn about factorising quadratics by looking at the opposite process, where two binomials are multiplied to get a quadratic. For example:

$$\begin{aligned}(x + 2)(x + 3) &= x^2 + 3x + 2x + 6 \\ &= x^2 + 5x + 6\end{aligned}$$

We see that the x^2 term in the quadratic is the product of the x -terms in each bracket. Similarly, the 6 in the quadratic is the product of the 2 and 3 in the brackets. Finally, the middle term is the sum of two terms.

So, how do we use this information to factorise the quadratic?

Let us start with factorising $x^2 + 5x + 6$ and see if we can decide upon some general rules. Firstly, write down the two brackets with an x in each bracket and space for the remaining terms.

$$(x \quad)(x \quad)$$

Next, decide upon the factors of 6. Since the 6 is positive, possible combinations are: 1 and 6, 2 and 3, -1 and -6 or -2 and -3

Therefore, we have four possibilities:

Option 1	Option 2	Option 3	Option 4
$(x + 1)(x + 6)$	$(x - 1)(x - 6)$	$(x + 2)(x + 3)$	$(x - 2)(x - 3)$

Next, we expand each set of brackets to see which option gives us the correct middle term.

Option 1	Option 2	Option 3	Option 4
$(x + 1)(x + 6)$	$(x - 1)(x - 6)$	$(x + 2)(x + 3)$	$(x - 2)(x - 3)$
$x^2 + 7x + 6$	$x^2 - 7x + 6$	<u>$x^2 + 5x + 6$</u>	$x^2 - 5x + 6$

We see that Option 3, $(x + 2)(x + 3)$, is the correct solution.

The process of factorising a quadratic is mostly trial and error but there are some strategies that can be used to ease the process.

1. Take out any common factor in the coefficients of the terms of the expression to obtain an expression of the form $ax^2 + bx + c$ where a , b and c have no common factors and a is positive.
2. Write down two brackets with an x in each bracket and space for the remaining terms: $(x \quad)(x \quad)$
3. Write down a set of factors for a and c .
4. Write down a set of options for the possible factors for the quadratic using the factors of a and c .
5. Expand all options to see which one gives you the correct middle term bx .

IMPORTANT!

If c is positive, then the factors of c must be either both positive or both negative. If c is negative, it means only one of the factors of c is negative, the other one being positive. Once you get an answer, always multiply out your brackets again just to make sure it really works.

VISIT:

The following video summarises how to factorise expressions and shows some examples.

► See video: 2DKX at www.everythingmaths.co.za

Worked example 13: Factorising a quadratic trinomial**QUESTION**

Factorise: $3x^2 + 2x - 1$.

SOLUTION

Step 1: Check that the quadratic is in required form $ax^2 + bx + c$

Step 2: Write down a set of factors for a and c

$$(x \quad)(x \quad)$$

The possible factors for a are: 1 and 3

The possible factors for c are: -1 and 1

Write down a set of options for the possible factors of the quadratic using the factors of a and c . Therefore, there are two possible options.

Option 1	Option 2
$(x - 1)(3x + 1)$	$(x + 1)(3x - 1)$
$3x^2 - 2x - 1$	$3x^2 + 2x - 1$

Step 3: Check that the solution is correct by multiplying the factors

$$\begin{aligned}(x + 1)(3x - 1) &= 3x^2 - x + 3x - 1 \\ &= 3x^2 + 2x - 1\end{aligned}$$

Step 4: Write the final answer

$$3x^2 + 2x - 1 = (x + 1)(3x - 1)$$

Exercise 1 – 8:

Factorise the following:

- | | | |
|---------------------------------|--------------------------------|---------------------------------|
| 1. $x^2 + 8x + 15$ | 2. $x^2 + 9x + 8$ | 3. $x^2 + 12x + 36$ |
| 4. $2h^2 + 5h - 3$ | 5. $3x^2 + 4x + 1$ | 6. $3s^2 + s - 10$ |
| 7. $x^2 - 2x - 15$ | 8. $x^2 + 2x - 3$ | 9. $x^2 + x - 20$ |
| 10. $x^2 - x - 20$ | 11. $2x^2 - 22x + 20$ | 12. $6a^2 + 14a + 8$ |
| 13. $6v^2 - 27v + 27$ | 14. $6g^2 - 15g - 9$ | 15. $3x^2 + 19x + 6$ |
| 16. $3x^2 + 17x - 6$ | 17. $7x^2 - 6x - 1$ | 18. $6x^2 - 15x - 9$ |
| 19. $a^2 - 7ab + 12b$ | 20. $3a^2 + 5ab - 12b^2$ | 21. $98x^4 + 14x^2 - 4$ |
| 22. $(x - 2)^2 - 7(x - 2) + 12$ | 23. $(a - 2)^2 - 4(a - 2) - 5$ | 24. $(y + 3)^2 - 3(y + 3) - 18$ |
| 25. $3(b^2 + 5b) + 12$ | 26. $6(a^2 + 3a) - 168$ | |

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

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|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2DKY | 2. 2DKZ | 3. 2DM2 | 4. 2DM3 | 5. 2DM4 | 6. 2DM5 |
| 7. 2DM6 | 8. 2DM7 | 9. 2DM8 | 10. 2DM9 | 11. 2DMB | 12. 2DMC |
| 13. 2DMD | 14. 2DMF | 15. 2DMG | 16. 2DMH | 17. 2DMJ | 18. 2DMK |
| 19. 2DMM | 20. 2DMN | 21. 2DMP | 22. 2DMQ | 23. 2DMR | 24. 2DMS |
| 25. 2DMT | 26. 2DMV | | | | |



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Sum and difference of two cubes

EMAP

We now look at two special results obtained from multiplying a binomial and a trinomial:

Sum of two cubes:

$$\begin{aligned}(x + y)(x^2 - xy + y^2) &= x(x^2 - xy + y^2) + y(x^2 - xy + y^2) \\&= [x(x^2) + x(-xy) + x(y^2)] + [y(x^2) + y(-xy) + y(y^2)] \\&= x^3 - x^2y + xy^2 + x^2y - xy^2 + y^3 \\&= x^3 + y^3\end{aligned}$$

Difference of two cubes:

$$\begin{aligned}(x - y)(x^2 + xy + y^2) &= x(x^2 + xy + y^2) - y(x^2 + xy + y^2) \\&= [x(x^2) + x(xy) + x(y^2)] - [y(x^2) + y(xy) + y(y^2)] \\&= x^3 + x^2y + xy^2 - x^2y - xy^2 - y^3 \\&= x^3 - y^3\end{aligned}$$

So we have seen that:

$$\begin{aligned}x^3 + y^3 &= (x + y)(x^2 - xy + y^2) \\x^3 - y^3 &= (x - y)(x^2 + xy + y^2)\end{aligned}$$

We use these two basic identities to factorise more complex examples.

Worked example 14: Factorising a difference of two cubes

QUESTION

Factorise: $a^3 - 1$.

SOLUTION

Step 1: Take the cube root of terms that are perfect cubes

We are working with the difference of two cubes. We know that $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$, so we need to identify x and y .

We start by noting that $\sqrt[3]{a^3} = a$ and $\sqrt[3]{1} = 1$. These give the terms in the first bracket. This also tells us that $x = a$ and $y = 1$.

Step 2: Find the three terms in the second bracket

We can replace x and y in the factorised form of the expression for the difference of two cubes with a and 1 . Doing so we get the second bracket:

$$(a^3 - 1) = (a - 1)(a^2 + a + 1)$$

Step 3: Expand the brackets to check that the expression has been correctly factorised

$$\begin{aligned}(a - 1)(a^2 + a + 1) &= a(a^2 + a + 1) - 1(a^2 + a + 1) \\ &= a^3 + a^2 + a - a^2 - a - 1 \\ &= a^3 - 1\end{aligned}$$

Worked example 15: Factorising a sum of two cubes

QUESTION

Factorise: $x^3 + 8$.

SOLUTION

Step 1: Take the cube root of terms that are perfect cubes

We are working with the sum of two cubes. We know that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$, so we need to identify x and y .

We start by noting that $\sqrt[3]{x^3} = x$ and $\sqrt[3]{8} = 2$. These give the terms in the first bracket. This also tells us that $x = x$ and $y = 2$.

Step 2: Find the three terms in the second bracket

We can replace x and y in the factorised form of the expression for the sum of two cubes with x and 2 . Doing so we get the second bracket:

$$(x^3 + 8) = (x + 2)(x^2 - 2x + 4)$$

Step 3: Expand the brackets to check that the expression has been correctly factorised

$$\begin{aligned}(x + 2)(x^2 - 2x + 4) &= x(x^2 - 2x + 4) + 2(x^2 - 2x + 4) \\ &= x^3 - 2x^2 + 4x + 2x^2 - 4x + 8 \\ &= x^3 + 8\end{aligned}$$

Worked example 16: Factorising a difference of two cubes

QUESTION

Factorise: $16y^3 - 432$.

SOLUTION

Step 1: Take out the common factor 16

$$16y^3 - 432 = 16(y^3 - 27)$$

Step 2: Take the cube root of terms that are perfect cubes

We are working with the difference of two cubes. We know that $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$, so we need to identify x and y .

We start by noting that $\sqrt[3]{y^3} = y$ and $\sqrt[3]{27} = 3$. These give the terms in the first bracket. This also tells us that $x = y$ and $y = 3$.

Step 3: Find the three terms in the second bracket

We can replace x and y in the factorised form of the expression for the difference of two cubes with y and 3. Doing so we get the second bracket:

$$16(y^3 - 27) = 16(y - 3)(y^2 + 3y + 9)$$

Step 4: Expand the brackets to check that the expression has been correctly factorised

$$\begin{aligned} 16(y - 3)(y^2 + 3y + 9) &= 16[(y(y^2 + 3y + 9) - 3(y^2 + 3y + 9))] \\ &= 16[y^3 + 3y^2 + 9y - 3y^2 - 9y - 27] \\ &= 16y^3 - 432 \end{aligned}$$

Worked example 17: Factorising a sum of two cubes

QUESTION

Factorise: $8t^3 + 125p^3$.

SOLUTION

Step 1: Take the cube root of terms that are perfect cubes

We are working with the sum of two cubes. We know that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$, so we need to identify x and y .

We start by noting that $\sqrt[3]{8t^3} = 2t$ and $\sqrt[3]{125p^3} = 5p$. These give the terms in the first bracket. This also tells us that $x = 2t$ and $y = 5p$.

Step 2: Find the three terms in the second bracket

We can replace x and y in the factorised form of the expression for the difference of two cubes with $2t$ and $5p$. Doing so we get the second bracket:

$$\begin{aligned}(8t^3 + 125p^3) &= (2t + 5p) \left[(2t)^2 - (2t)(5p) + (5p)^2 \right] \\ &= (2t + 5p) (4t^2 - 10tp + 25p^2)\end{aligned}$$

Step 3: Expand the brackets to check that the expression has been correctly factorised

$$\begin{aligned}(2t + 5p) (4t^2 - 10tp + 25p^2) &= 2t (4t^2 - 10tp + 25p^2) + 5p (4t^2 - 10tp + 25p^2) \\ &= 8t^3 - 20pt^2 + 50p^2t + 20pt^2 - 50p^2t + 125p^3 \\ &= 8t^3 + 125p^3\end{aligned}$$

Exercise 1 – 9:

Factorise:

- | | | |
|---|---|----------------------------|
| 1. $w^3 - 8$ | 2. $g^3 + 64$ | 3. $h^3 + 1$ |
| 4. $x^3 + 8$ | 5. $27 - m^3$ | 6. $2x^3 - 2y^3$ |
| 7. $3k^3 + 81q^3$ | 8. $64t^3 - 1$ | 9. $64x^2 - 1$ |
| 10. $125x^3 + 1$ | 11. $25x^3 + 1$ | 12. $z - 125z^4$ |
| 13. $8m^6 + n^9$ | 14. $216n^3 - k^3$ | 15. $125s^3 + d^3$ |
| 16. $8k^3 + r^3$ | 17. $8j^3k^3l^3 - b^3$ | 18. $27x^3y^3 + w^3$ |
| 19. $128m^3 + 2f^3$ | 20. $p^{15} - \frac{1}{8}y^{12}$ | 21. $\frac{27}{t^3} - s^3$ |
| 22. $\frac{1}{64q^3} - h^3$ | 23. $72g^3 + \frac{1}{3}v^3$ | 24. $1 - (x - y)^3$ |
| 25. $h^4(8g^6 + h^3) - (8g^6 + h^3)$ | 26. $x(125w^3 - h^3) + y(125w^3 - h^3)$ | |
| 27. $x^2(27p^3 + w^3) - 5x(27p^3 + w^3) - 6(27p^3 + w^3)$ | | |

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2DMW | 2. 2DMX | 3. 2DMY | 4. 2DMZ | 5. 2DN2 | 6. 2DN3 | 7. 2DN4 |
| 8. 2DN5 | 9. 2DN6 | 10. 2DN7 | 11. 2DN8 | 12. 2DN9 | 13. 2DNB | 14. 2DNC |
| 15. 2DND | 16. 2DNF | 17. 2DNG | 18. 2DNH | 19. 2DNJ | 20. 2DNK | 21. 2DNM |
| 22. 2DNN | 23. 2DNP | 24. 2DNQ | 25. 2DNR | 26. 2DNS | 27. 2DNT | |



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We have studied procedures for working with fractions in earlier grades.

1. $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \quad (b \neq 0; d \neq 0)$
2. $\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b} \quad (b \neq 0)$
3. $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc} \quad (b \neq 0; c \neq 0; d \neq 0)$

Note: dividing by a fraction is the same as multiplying by the reciprocal of the fraction.

In some cases of simplifying an algebraic expression, the expression will be a fraction. For example,

$$\frac{x^2 + 3x}{x + 3}$$

has a quadratic binomial in the numerator and a linear binomial in the denominator. We have to apply the different factorisation methods in order to factorise the numerator and the denominator before we can simplify the expression.

$$\begin{aligned} \frac{x^2 + 3x}{x + 3} &= \frac{x(x + 3)}{x + 3} \\ &= x \quad (x \neq -3) \end{aligned}$$

If $x = -3$ then the denominator, $x + 3 = 0$ and the fraction is undefined.

VISIT:

This video shows some examples of simplifying fractions.

▶ See video: [2DNV](https://www.youtube.com/watch?v=2DNV) at www.everythingmaths.co.za

Worked example 18: Simplifying fractions

QUESTION

Simplify:

$$\frac{ax - b + x - ab}{ax^2 - abx}, \quad (x \neq 0; x \neq b)$$

SOLUTION

Step 1: Use grouping to factorise the numerator and take out the common factor ax in the denominator

$$\frac{(ax - ab) + (x - b)}{ax^2 - abx} = \frac{a(x - b) + (x - b)}{ax(x - b)}$$

Step 2: Take out common factor $(x - b)$ in the numerator

$$= \frac{(x - b)(a + 1)}{ax(x - b)}$$

Step 3: Cancel the common factor in the numerator and the denominator to give the final answer

$$= \frac{a+1}{ax}$$

Worked example 19: Simplifying fractions

QUESTION

Simplify:

$$\frac{x^2 - x - 2}{x^2 - 4} \div \frac{x^2 + x}{x^2 + 2x}, \quad (x \neq 0; x \neq \pm 2)$$

SOLUTION

Step 1: Factorise the numerator and denominator

$$= \frac{(x+1)(x-2)}{(x+2)(x-2)} \div \frac{x(x+1)}{x(x+2)}$$

Step 2: Change the division sign and multiply by the reciprocal

$$= \frac{(x+1)(x-2)}{(x+2)(x-2)} \times \frac{x(x+2)}{x(x+1)}$$

Step 3: Write the final answer

$$= 1$$

Worked example 20: Simplifying fractions

QUESTION

Simplify:

$$\frac{x-2}{x^2-4} + \frac{x^2}{x-2} - \frac{x^3+x-4}{x^2-4}, \quad (x \neq \pm 2)$$

SOLUTION

Step 1: Factorise the denominators

$$\frac{x-2}{(x+2)(x-2)} + \frac{x^2}{x-2} - \frac{x^3+x-4}{(x+2)(x-2)}$$

Step 2: Make all denominators the same so that we can add or subtract the fractions

The lowest common denominator is $(x - 2)(x + 2)$.

$$\frac{x - 2}{(x + 2)(x - 2)} + \frac{(x^2)(x + 2)}{(x + 2)(x - 2)} - \frac{x^3 + x - 4}{(x + 2)(x - 2)}$$

Step 3: Write as one fraction

$$\frac{x - 2 + (x^2)(x + 2) - (x^3 + x - 4)}{(x + 2)(x - 2)}$$

Step 4: Simplify

$$\frac{x - 2 + x^3 + 2x^2 - x^3 - x + 4}{(x + 2)(x - 2)} = \frac{2x^2 + 2}{(x + 2)(x - 2)}$$

Step 5: Take out the common factor and write the final answer

$$\frac{2(x^2 + 1)}{(x + 2)(x - 2)}$$

Worked example 21: Simplifying fractions**QUESTION**

Simplify:

$$\frac{2}{x^2 - x} + \frac{x^2 + x + 1}{x^3 - 1} - \frac{x}{x^2 - 1}, \quad (x \neq 0; x \neq \pm 1)$$

SOLUTION**Step 1: Factorise the numerator and denominator**

$$\frac{2}{x(x - 1)} + \frac{(x^2 + x + 1)}{(x - 1)(x^2 + x + 1)} - \frac{x}{(x - 1)(x + 1)}$$

Step 2: Simplify and find the common denominator

$$\frac{2(x + 1) + x(x + 1) - x^2}{x(x - 1)(x + 1)}$$

Step 3: Write the final answer

$$\frac{2x + 2 + x^2 + x - x^2}{x(x - 1)(x + 1)} = \frac{3x + 2}{x(x - 1)(x + 1)}$$

1. Simplify (assume all denominators are non-zero):

a) $\frac{3a}{15}$	b) $\frac{2a+10}{4}$	c) $\frac{5a+20}{a+4}$
d) $\frac{a^2-4a}{a-4}$	e) $\frac{3a^2-9a}{2a-6}$	f) $\frac{9a+27}{9a+18}$
g) $\frac{6ab+2a}{2b}$	h) $\frac{16x^2y-8xy}{12x-6}$	i) $\frac{4xyp-8xp}{t^2-s^2}$
j) $\frac{9x^2-16}{6x-8}$	k) $\frac{b^2-81a^2}{18a-2b}$	l) $\frac{12xy}{s^2-2st+t^2}$
m) $\frac{x^2-2x-15}{5x-25}$	n) $\frac{x^2+8x+15}{a^2-4ab-12b^2}$	o) $\frac{x^3-27}{6a^2-7a-3}$
p) $\frac{a^2+6a-16}{a^3-8}$	q) $\frac{a^2+4ab+4b^2}{qz+qr+16z+16r}$	r) $\frac{3ab+b}{z-q}$
s) $\frac{2x^2-x-1}{x^3-x}$	t) $\frac{f^2a-fa^2}{f-a}$	u) $\frac{pz-pq+5z-5q}{z-q}$
v) $\frac{hx-hg+13x-13g}{x-g}$	w) $\frac{f^2a-fa^2}{f-a}$	

2. Simplify (assume all denominators are non-zero):

a) $\frac{b^2+10b+21}{3(b^2-9)} \div \frac{2b^2+14b}{30b^2-90b}$	b) $\frac{x^2+17x+70}{5(x^2-100)} \div \frac{3x^2+21x}{45x^2-450x}$
c) $\frac{z^2+17z+66}{3(z^2-121)} \div \frac{2z^2+12z}{24z^2-264z}$	d) $\frac{3a+9}{14} \div \frac{7a+21}{a+3}$
e) $\frac{a^2-5a}{2a+10} \times \frac{4a}{3a+15}$	f) $\frac{3xp+4p}{8p} \div \frac{12p^2}{3x+4}$
g) $\frac{24a-8}{12} \div \frac{9a-3}{6}$	h) $\frac{a^2+2a}{5} \div \frac{2a+4}{20}$
i) $\frac{p^2+pq}{7p} \times \frac{21q}{8p+8q}$	j) $\frac{5ab-15b}{4a-12} \div \frac{6b^2}{a+b}$
k) $\frac{16-x^2}{x^2-x-12} \times \frac{x+3}{x+4}$	l) $\frac{a^3+b^3}{a^3} \times \frac{5a+5b}{a^2+2ab+b^2}$
m) $\frac{a-4}{a+5a+4} \times \frac{a^2+2a+1}{a^2-3a-4}$	n) $\frac{3x+2}{x^2-6x+8} \times \frac{x-2}{3x^2+8x+4}$
o) $\frac{a^2-2a+8}{a^2+6a+8} \times \frac{a^2+a-12}{3} - \frac{3}{2}$	p) $\frac{4x^2-1}{3x^2+10x+3} \div \frac{6x^2+5x+1}{4x^2+7x-3} \times \frac{9x^2+6x+1}{8x^2-6x+1}$
q) $\frac{x+4}{3} - \frac{x-2}{2}$	r) $\frac{p^3+q^3}{p^2} \times \frac{3p-3q}{p^2-q^2}$

3. Simplify (assume all denominators are non-zero):

a) $\frac{x-3}{3} - \frac{x+5}{4}$	b) $\frac{2x-4}{9} - \frac{x-3}{4} + 1$
c) $1 + \frac{3x-4}{4} - \frac{x+2}{3}$	d) $\frac{11}{a+11} + \frac{8}{a-8}$
e) $\frac{12}{x-12} - \frac{6}{x-6}$	f) $\frac{12}{r+12} + \frac{8}{r-8}$

$$g) \frac{2}{xy} + \frac{4}{xz} + \frac{3}{yz}$$

$$i) \frac{k+2}{k^2+2} - \frac{1}{k+2}$$

$$k) \frac{3}{p^2-4} + \frac{2}{(p-2)^2}$$

$$m) \frac{1}{m+n} + \frac{3mn}{m^3+n^3}$$

$$o) \frac{x^2-1}{3} \times \frac{1}{x-1} - \frac{1}{2}$$

$$q) \frac{1}{(x-1)^2} - \frac{2x}{x^3-1}$$

$$s) \frac{x^2-3x+9}{x^3+27} + \frac{x-2}{x^2+4x+3} - \frac{1}{x-2}$$

$$h) \frac{5}{t-2} - \frac{1}{t-3}$$

$$j) \frac{t+2}{3q} + \frac{t+1}{2q}$$

$$l) \frac{x}{x+y} + \frac{x^2}{y^2-x^2}$$

$$n) \frac{h}{h^3-f^3} - \frac{1}{h^2+hf+f^2}$$

$$p) \frac{x^2-2x+1}{(x-1)^3} - \frac{x^2+x+1}{x^3-1}$$

$$r) \frac{t^2+2t-8}{t^2+t-6} + \frac{1}{t^2-9} + \frac{t+1}{t-3}$$

$$t) \frac{1}{a^2-4ab+4b^2} + \frac{a^2+2ab+b^2}{a^3-8b^3} - \frac{1}{a^2-4b^2}$$

4. What are the restrictions in the following:

$$a) \frac{1}{x-2}$$

$$b) \frac{3x-9}{4x+4}$$

$$c) \frac{3}{x} - \frac{1}{x^2-1}$$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2DNW	1b. 2DNX	1c. 2DNY	1d. 2DNZ	1e. 2DP2	1f. 2DP3	1g. 2DP4	1h. 2DP5
1i. 2DP6	1j. 2DP7	1k. 2DP8	1l. 2DP9	1m. 2DPB	1n. 2DPC	1o. 2DPD	1p. 2DPF
1q. 2DPG	1r. 2DPH	1s. 2DPJ	1t. 2DPK	1u. 2DPM	1v. 2DPN	1w. 2DPP	2a. 2DPQ
2b. 2DPR	2c. 2DPS	2d. 2DPT	2e. 2DPV	2f. 2DPW	2g. 2DPX	2h. 2DPY	2i. 2DPZ
2j. 2DQ2	2k. 2DQ3	2l. 2DQ4	2m. 2DQ5	2n. 2DQ6	2o. 2DQ7	2p. 2DQ8	2q. 2DQ9
2r. 2DQB	3a. 2DQC	3b. 2DQD	3c. 2DQF	3d. 2DQG	3e. 2DQH	3f. 2DQJ	3g. 2DQK
3h. 2DQM	3i. 2DQN	3j. 2DQP	3k. 2DQQ	3l. 2DQR	3m. 2DQS	3n. 2DQT	3o. 2DQV
3p. 2DQW	3q. 2DQX	3r. 2DQY	3s. 2DQZ	3t. 2DR2	4a. 2DR3	4b. 2DR4	4c. 2DR5



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1.9 Chapter summary

EMAR

► See presentation: [2DR6](#) at www.everythingmaths.co.za

- $\mathbb{N}$: natural numbers are $\{1; 2; 3; \dots\}$
- $\mathbb{N}_0$: whole numbers are $\{0; 1; 2; 3; \dots\}$
- $\mathbb{Z}$: integers are $\{\dots; -3; -2; -1; 0; 1; 2; 3; \dots\}$
- A rational number is any number that can be written as $\frac{a}{b}$ where a and b are integers and $b \neq 0$.
- The following are rational numbers:
 - Fractions with both numerator and denominator as integers
 - Integers
 - Decimal numbers that terminate
 - Decimal numbers that recur (repeat)

- Irrational numbers are numbers that cannot be written as a fraction with the numerator and denominator as integers.
- If the n^{th} root of a number cannot be simplified to a rational number, it is called a surd.
- If a and b are positive whole numbers, and $a < b$, then $\sqrt[n]{a} < \sqrt[n]{b}$.
- A binomial is an expression with two terms.
- The product of two identical binomials is known as the square of the binomial.
- We get the difference of two squares when we multiply $(ax + b)(ax - b)$
- Factorising is the opposite process of expanding the brackets.
- The product of a binomial and a trinomial is:

$$(A + B)(C + D + E) = A(C + D + E) + B(C + D + E)$$

- Taking out a common factor is the basic factorisation method.
- We often need to use grouping to factorise polynomials.
- To factorise a quadratic we find the two binomials that were multiplied together to give the quadratic.
- The sum of two cubes can be factorised as:

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

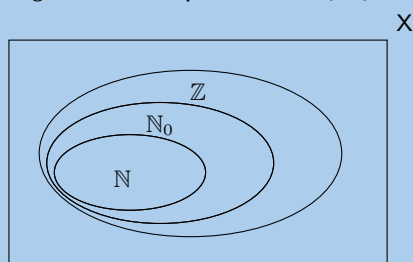
- The difference of two cubes can be factorised as:

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

- We can simplify fractions by incorporating the methods we have learnt to factorise expressions.
- Only factors can be cancelled out in fractions, never terms.
- To add or subtract fractions, the denominators of all the fractions must be the same.

End of chapter Exercise 1 – 11:

1. The figure here shows the Venn diagram for the special sets $\mathbb{N}$, $\mathbb{N}_0$ and $\mathbb{Z}$.



- a) Where does the number 2,13 belong in the diagram?
 - b) In the following list, there are two false statements and one true statement. Which of the statements is **true**?
 - Every natural number is an integer.
 - Every whole number is a natural number.
 - There are fractions in the integers.
2. State whether the following numbers are real, non-real or undefined.

- a) $-\sqrt{-5}$ b) $\frac{\sqrt{8}}{0}$ c) $-\sqrt{15}$ d) $-\sqrt{7}$ e) $\sqrt{-1}$ f) $\sqrt{2}$

3. State whether each of the following numbers are rational or irrational.

- a) $\sqrt[3]{4}$ b) 45π c) $\sqrt{9}$ d) $\sqrt[3]{8}$

4. If a is an integer, b is an integer and c is irrational, which of the following are rational numbers?

- a) $\frac{-b}{a}$ b) $c \div c$ c) $\frac{a}{c}$ d) $\frac{1}{c}$

5. Consider the following list of numbers:

$$\sqrt[3]{26}; \frac{3}{2}; \sqrt{-24}; \sqrt{39}; 7,1\dot{1}; \pi^2; \frac{\pi}{2}; 7,12; -\sqrt{24}; \frac{\sqrt{2}}{0}; 3\pi; \sqrt{78}; 9; \pi$$

- Which of the numbers are non-real numbers?
- Without using a calculator, rank all the real numbers in ascending order.
- Which of the numbers are irrational numbers?
- Which of the numbers are rational numbers?
- Which of the numbers are integers?
- Which of the numbers are undefined?

6. Write each decimal as a simple fraction.

- a) 0,12 b) 0,006 c) $4,\overline{14}$ d) 1,59
e) $12,27\dot{7}$ f) $0,8\dot{2}$ g) $7,\overline{36}$

7. Show that the decimal $3,21\dot{1}\dot{8}$ is a rational number.

8. Write the following fractions as decimal numbers:

- a) $\frac{1}{18}$ b) $1\frac{1}{2}$

9. Express $0,\overline{78}$ as a fraction $\frac{a}{b}$ where $a, b \in \mathbb{Z}$ (show all working).

10. For each of the following numbers:

- write the next three digits;
- state whether the number is rational or irrational.

a) 1,11235...

b) $1,\dot{1}$

11. Write the following rational numbers to 2 decimal places.

- a) $\frac{1}{2}$ b) 1 c) $0,1111\overline{1}$ d) $0,99999\overline{1}$

12. Round off the following irrational numbers to 3 decimal places.

- a) 3,141592654... b) 1,618033989...
c) 1,41421356... d) 2,71828182845904523536...

13. Round off the number 1523,00195593 to 4 decimal places.

14. Round off the number 1982,94028996 to 6 decimal places.

15. Round off the number 101,52378984 to 4 decimal places.

16. Use your calculator and write the following irrational numbers to 3 decimal places.

- a) $\sqrt{2}$ b) $\sqrt{3}$ c) $\sqrt{5}$ d) $\sqrt{6}$

17. Use your calculator (where necessary) and write the following numbers to 5 decimal places. State whether the numbers are irrational or rational.

- a) $\sqrt{8}$ b) $\sqrt{768}$ c) $\sqrt{0,49}$ d) $\sqrt{0,0016}$ e) $\sqrt{0,25}$
f) $\sqrt{36}$ g) $\sqrt{1960}$ h) $\sqrt{0,0036}$ i) $-8\sqrt{0,04}$ j) $5\sqrt{80}$

18. Round off:

- $\frac{\sqrt{2}}{2}$ to the nearest 2 decimal places.
- $\sqrt{14}$ to the nearest 3 decimal places.

19. Write the following irrational numbers to 3 decimal places and then write each one as a rational number to get an approximation of the irrational number.
a) 3,141592654... b) 1,618033989...
c) 1,41421356... d) 2,71828182845904523536...
20. Determine between which two consecutive integers the following irrational numbers lie, without using a calculator.
a) $\sqrt{5}$ b) $\sqrt{10}$ c) $\sqrt{20}$ d) $\sqrt{30}$ e) $\sqrt[3]{5}$ f) $\sqrt[3]{10}$
g) $\sqrt[3]{20}$ h) $\sqrt[3]{30}$ i) $\sqrt{90}$ j) $\sqrt{72}$ k) $\sqrt[3]{58}$ l) $\sqrt[3]{118}$
21. Estimate the following surds to the nearest 1 decimal place, without using a calculator.
a) $\sqrt{14}$ b) $\sqrt{110}$ c) $\sqrt{48}$ d) $\sqrt{57}$
22. Expand the following products:
a) $(a + 5)^2$ b) $(n + 12)^2$ c) $(d - 4)^2$
d) $(7w + 2)(7w - 2)$ e) $(12q + 1)(12q - 1)$ f) $-(-x - 2)(x + 2)$
g) $(5k - 4)(5k + 4)$ h) $(5f + 4)(2f + 2)$ i) $(3n + 6)(6n + 5)$
j) $(2g + 6)(g + 6)$ k) $(4y + 1)(4y + 8)$ l) $(d - 3)(7d + 2)$
m) $(6z - 4)(z - 2)$ n) $(5w - 11)^2$ o) $(5s - 1)^2$
p) $(3d - 8)^2$ q) $5f^2(3f + 5) + 7f(3f^2 + 7)$ r) $8d(4d^3 + 2) + 6d^2(7d^2 + 4)$
s) $5x^2(2x + 2) + 7x(7x^2 + 7)$
23. Expand the following:
a) $(y^4 + 3y^2 + y)(y + 1)(y - 2)$ b) $(x + 1)^2 - (x - 1)^2$
c) $(x^2 + 2x + 1)(x^2 - 2x + 1)$ d) $(4a - 3b)(16a^2 + 12ab + 9b^2)$
e) $2(x + 3y)(x^2 - xy - y^2)$ f) $(3a - 5b)(3a + 5b)(a^2 + ab - b^2)$
g) $\left(y - \frac{1}{y}\right)\left(y + \frac{1}{y}\right)$ h) $\left(\frac{a}{3} - \frac{3}{a}\right)\left(\frac{a}{3} + \frac{3}{a}\right)$
i) $\frac{1}{3}(12x - 9y) + \frac{1}{6}(12x + 18y)$ j) $(x + 2)(x - 2) - (x + 2)^2$
24. What is the value of e in $(x - 4)(x + e) = x^2 - 16$?
25. In $(x + 2)(x + k) = x^2 + bx + c$:
a) For which of these values of k will b be positive?
 $-6 ; -1 ; 0 ; 1 ; 6$
b) For which of these values of k will c be positive?
 $-6 ; -1 ; 0 ; 1 ; 6$
c) For what values of k will c be positive?
d) For what values of k will b be positive?
26. Answer the following:
a) Expand: $\left(3a - \frac{1}{2a}\right)^2$
b) Expand: $\left(3a - \frac{1}{2a}\right)\left(9a^2 + \frac{3}{2} + \frac{1}{4a^2}\right)$
c) Given that $3a - \frac{1}{2a} = 7$, determine the value of $27a^3 - \frac{1}{8a^3}$ without solving for a .

27. Solve by factorising:

- a) $17^2 - 15^2$ b) $13^2 - 12^2$ c) $120045^2 - 120035^2$ d) $26^2 - 24^2$

28. Represent the following as a product of its prime factors:

- a) 143 b) 168 c) 899 d) 99 e) 1599

29. Factorise:

- a) $a^2 - 9$ b) $9b^2 - 81$
c) $m^2 - \frac{1}{9}$ d) $5 - 5a^2b^6$
e) $16ba^4 - 81b$ f) $a^2 - 10a + 25$
g) $16b^2 + 56b + 49$ h) $-4b^2 - 144b^8 + 48b^5$
i) $16 - x^4$ j) $7x^2 - 14x + 7xy - 14y$
k) $y^2 - 7y - 30$ l) $1 - x - x^2 + x^3$
m) $-3(1 - p^2) + p + 1$ n) $x^2 - 2x + 1 - y^4$
o) $4b(x^3 - 1) + x(1 - x^3)$ p) $3m(v - 7) + 19(-7 + v)$
q) $3f(z + 3) + 19(3 + z)$ r) $3p^3 - \frac{1}{9}$
s) $8x^6 - 125y^9$ t) $(2 + p)^3 - 8(p + 1)^3$
u) $\frac{1}{3}a^3 - a^2b + 2a^2b - 6ab^2 + 3ab^2 - 9b^3$ v) $6a^2 - 17a + 5$
w) $s^2 + 2s - 15$ x) $16v + 24h + 2j^5v + 3j^5h$
y) $18h - 45g + 2m^3h - 5m^3g$ z) $63d - 18s + 7u^2d - 2u^2s$

30. Factorise the following:

- a) $6a^2 + 14a + 8$ b) $6g^2 - 15g - 9$
c) $125g^3 - r^3$ d) $8r^3 + z^3$
e) $14m - 4n + 7jm - 2jn$ f) $25d - 15m + 5yd - 3ym$
g) $g^3 - 27$ h) $z^3 + 125$
i) $b^2 - (3a - 2b)^2$ j) $9y^2 - (4x + 2y)^2$
k) $16x^6 - 3y^8$ l) $\frac{1}{6}a^2 - 24b^4$
m) $4(a - 3) - 81x^2(a - 3)$ n) $(2 + b)^2 - 11(2 + b) - 12$
o) $2x^2 + 7xy + 5y^2$ p) $x^2 - 2xy - 15y^2$
q) $4x^4 + 11x^2 + 6$ r) $6x^4 - 38x^2 + 40$
s) $9a^2x + 9a^2y + 27a^2 - b^2x - b^2y - 3b^2$ t) $2(2y^2 - 5y) - 24$
u) $\frac{1}{2}x^3 - \frac{9}{2}x - 2x^2 + 18$ v) $27r^3s^3 - 1$
w) $\frac{1}{125h^3} + r^3$ x) $j(64n^3 - b^3) + k(64n^3 - b^3)$

31. Simplify the following:

- a) $(a - 2)^2 - a(a + 4)$ b) $(5a - 4b)(25a^2 + 20ab + 16b^2)$
c) $(2m - 3)(4m^2 + 9)(2m + 3)$ d) $(a + 2b - c)(a + 2b + c)$

$$e) \frac{m^2 + 11m + 18}{4(m^2 - 4)} \div \frac{3m^2 + 27m}{24m^2 - 48m}$$

$$g) \frac{4 - b^2}{3b - 6}$$

$$i) \frac{x^2 - 5x - 14}{3x + 6}$$

$$k) \frac{a - 2}{a^2 + 4a + 3} \div \frac{(a - 1)(a + 1)}{a - 1} \times \frac{a^2 - 2a - 15}{a - 2}$$

$$m) 2 \div \frac{a + b}{a + 2b} \times \frac{b^2 - ba - 6a^2}{a^2 - 4b^2} \times \frac{a^2 - b - 2b^2}{3a - b}$$

$$o) \frac{ny + nq + 8y + 8q}{y + q}$$

$$q) \frac{2}{x} + \frac{x}{2} - \frac{2x}{3}$$

$$s) \frac{x + 2}{2x^3} + 16$$

$$u) \frac{1}{2}x + \frac{x - 2}{3} + 4$$

$$w) \frac{b^2 + 6b + 9}{b^2 - 9} + \frac{b^2 - 6b + 8}{(b - 2)(b + 3)} + \frac{1}{b + 3}$$

$$y) \frac{12}{z + 12} + \frac{5}{z - 5}$$

$$f) \frac{t^2 + 9t + 18}{5(t^2 - 9)} \div \frac{4t^2 + 24t}{100t^2 - 300t}$$

$$h) \frac{x^2 + 2x + 4}{x^3 - 8}$$

$$j) \frac{d^2 + 23d + 132}{5(d^2 - 121)} \div \frac{4d^2 + 48d}{100d^2 - 1100d}$$

$$l) \frac{a + 6}{a^2 + 12a + 11} \times \frac{a^2 + 14a + 33}{a + 3} \div \frac{a^3 + 216}{a + 1}$$

$$n) \frac{st + sb + 31t + 31b}{t + b}$$

$$p) \frac{p^2 - q^2}{p} \div \frac{p + q}{p^2 - pq}$$

$$r) \frac{1}{a + 7} - \frac{a + 7}{a^2 - 49}$$

$$t) \frac{1 - 2a}{4a^2 - 1} - \frac{a - 1}{2a^2 - 3a + 1} - \frac{1}{1 - a}$$

$$v) \frac{1}{x^2 + 2x} + \frac{4x^2 - x - 3}{x^2 + 2x - 3}$$

$$x) \frac{x^2 + 2x}{x^2 + x + 6} \times \frac{x^2 + 2x + 1}{x^2 + 3x + 2}$$

$$z) \frac{11}{w - 11} - \frac{4}{w - 4}$$

32. Show that $(2x - 1)^2 - (x - 3)^2$ can be simplified to $(x + 2)(3x - 4)$.

33. What must be added to $x^2 - x + 4$ to make it equal to $(x + 2)^2$?

34. Evaluate $\frac{x^3 + 1}{x^2 - x + 1}$ if $x = 7,85$ without using a calculator. Show your work.

35. With what expression must $(a - 2b)$ be multiplied to get a product of $(a^3 - 8b^3)$?

36. With what expression must $27x^3 + 1$ be divided to get a quotient of $3x + 1$?

37. What are the restrictions on the following?

$$a) \frac{4}{3x^2 + 2x - 1}$$

$$b) \frac{a}{3(b - a) + ab - a^2}$$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2DR7	2a. 2DR8	2b. 2DR9	2c. 2DRB	2d. 2DRC	2e. 2DRD	2f. 2DRF
3a. 2DRG	3b. 2DRH	3c. 2DRJ	3d. 2DRK	4. 2DRM	5. 2DRN	6a. 2DRP
6b. 2DRQ	6c. 2DRR	6d. 2DRS	6e. 2DRT	6f. 2DRV	6g. 2DRW	7. 2DRX
8a. 2DRY	8b. 2DRZ	9. 2DS2	10a. 2DS3	10b. 2DS4	11a. 2DS5	11b. 2DS6
11c. 2DS7	11d. 2DS8	12a. 2DS9	12b. 2DSB	12c. 2DSC	12d. 2DSD	13. 2DSF
14. 2DSG	15. 2DSH	16a. 2DSJ	16b. 2DSK	16c. 2DSM	16d. 2DSN	17a. 2DSP
17b. 2DSQ	17c. 2DSR	17d. 2DSS	17e. 2DST	17f. 2DSV	17g. 2DSW	17h. 2DSX
17i. 2DSY	17j. 2DSZ	18a. 2DT2	18b. 2DT3	19a. 2DT4	19b. 2DT5	19c. 2DT6
19d. 2DT7	20a. 2DT8	20b. 2DT9	20c. 2DTB	20d. 2DTC	20e. 2DTD	20f. 2DTF
20g. 2DTG	20h. 2DTH	20i. 2DTJ	20j. 2DTK	20k. 2DTM	20l. 2DTN	21a. 2DTP
21b. 2DTQ	21c. 2DTR	21d. 2DTS	22a. 2DTT	22b. 2DTW	22c. 2DTX	22d. 2DTY
22e. 2DTZ	22f. 2DV2	22g. 2DV3	22h. 2DV4	22i. 2DV5	22j. 2DV6	22k. 2DV7
22l. 2DV8	22m. 2DV9	22n. 2DVB	22o. 2DVC	22p. 2DVD	22q. 2DVF	22r. 2DVG
22s. 2DVH	23a. 2DVJ	23b. 2DVK	23c. 2DVM	23d. 2DVN	23e. 2DVP	23f. 2DVQ
23g. 2DVR	23h. 2DVS	23i. 2DVT	23j. 2DVV	24. 2DVW	25. 2DVX	26. 2DVY

27a. 2DVZ	27b. 2DW2	27c. 2DW3	27d. 2DW4	28a. 2DW5	28b. 2DW6	28c. 2DW7
28d. 2DW8	28e. 2DW9	29a. 2DWB	29b. 2DWC	29c. 2DWD	29d. 2DWF	29e. 2DWG
29f. 2DWH	29g. 2DWJ	29h. 2DWK	29i. 2DWM	29j. 2DWN	29k. 2DWP	29l. 2DWQ
29m. 2DWR	29n. 2DWS	29o. 2DWT	29p. 2DWV	29q. 2DWW	29r. 2DWX	29s. 2DWY
29t. 2DWZ	29u. 2DX2	29v. 2DX3	29w. 2DX4	29x. 2DX5	29y. 2DX6	29z. 2DX7
30a. 2DX8	30b. 2DX9	30c. 2DXB	30d. 2DXC	30e. 2DXD	30f. 2DXF	30g. 2DXG
30h. 2DXH	30i. 2DXJ	30j. 2DXK	30k. 2DXM	30l. 2DXN	30m. 2DXP	30n. 2DXQ
30o. 2DXR	30p. 2DXS	30q. 2DXT	30r. 2DXV	30s. 2DXW	30t. 2DXX	30u. 2DXY
30v. 2DXZ	30w. 2DY2	30x. 2DY3	31a. 2DY4	31b. 2DY5	31c. 2DY6	31d. 2DY7
31e. 2DY8	31f. 2DY9	31g. 2DYB	31h. 2DYC	31i. 2DYD	31j. 2DYF	31k. 2DYG
31l. 2DYH	31m. 2DYJ	31n. 2DYK	31o. 2DYM	31p. 2DYN	31q. 2DYP	31r. 2DYQ
31s. 2DYR	31t. 2DYS	31u. 2DYT	31v. 2DYV	31w. 2DYW	31x. 2DYX	31y. 2DYY
31z. 2DYZ	32. 2DZ2	33. 2DZ3	34. 2DZ4	35. 2DZ5	36. 2DZ6	37a. 2DZ7
37b. 2DZ8						



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Exponents

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2.1 Introduction

EMAS

Exponential notation is a short way of writing the same number multiplied by itself many times. This is very useful in everyday life. You may have heard someone describe the size of an area in *square* metres or *square* kilometres. For example, the largest radio telescope in the world is being built in South Africa. The telescope is called the *square kilometre* array, or SKA. This is because the telescope will occupy an area of 1 kilometre by 1 kilometre or 1 kilometre squared.

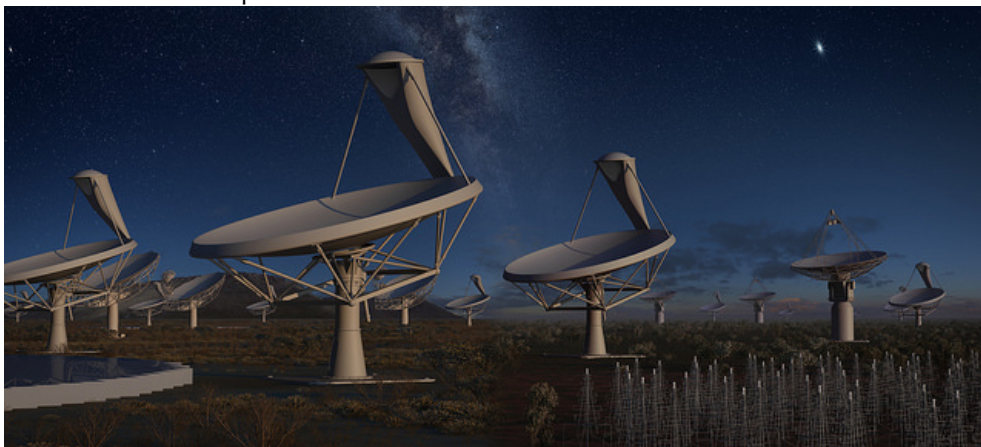


Figure 2.1: Antennas from the Square Kilometre Array (artist's concept).

Exponents are also very useful to describe very large and very small numbers. For example, the SKA will be detecting incredibly weak signals from objects which are so far away that to write out the strength of the signal or the number of kilometres away in full would be impractical. Outside of astronomy, exponents are used by many other professions such as computer programmers, engineers, economists, financial analysts, biologists and demographers.

VISIT:

If you want to know more about how exponents are used then take a look at the following presentation.

▶ See presentation: [2DZ9](#) at www.everythingmaths.co.za

You have already been introduced to exponents and exponent laws in previous grades. Remember that exponents can also be called indices or powers. Exponential notation is as follows:

$$\text{base} \leftarrow a^n \rightarrow \text{exponent or index}$$

For any real number a and natural number n , we can write a multiplied by itself n times as: a^n .

Remember the following identities:

1. $a^n = a \times a \times a \times \cdots \times a$ (n times) ($a \in \mathbb{R}, n \in \mathbb{N}$)
2. $a^0 = 1$ ($a \neq 0$ because 0^0 is undefined)
3. $a^{-n} = \frac{1}{a^n}$ ($a \neq 0$ because $\frac{1}{0}$ is undefined)
4. Similarly, $\frac{1}{a^{-n}} = a^n$

VISIT:

Interested in finding out why raising a real number to the power of zero is one? Try work it out for yourself. If you get stuck, you can see an example of how to show this is true at this link.

▶ See video: [2DZB](#) at www.everythingmaths.co.za

Look at the following examples to see these identities in action:

1. $3 \times 3 = 3^2 = 9$
2. $5 \times 5 \times 5 \times 5 = 5^4$
3. $p \times p \times p = p^3$
4. $(3^x)^0 = 1$
5. $2^{-4} = \frac{1}{2^4} = \frac{1}{16}$
6. $\frac{1}{5^{-x}} = 5^x$

NOTE:

If your final answer is easier to work out without a calculator, then write it out in full - not in exponential notation, as in examples 1 and 5.

NOTE:

It is convention to write your final answer with positive exponents.

In this chapter, we will revise the exponent laws and use these laws to simplify and solve more complex expressions and equations.

VISIT:

To revise what exponents are you can watch the following video.

▶ See video: [2DZC](#) at www.everythingmaths.co.za

2.2 Revision of exponent laws

EMAT

There are several laws we can use to make working with exponential numbers easier. Some of these laws might have been done in earlier grades, but we list all the laws here for easy reference:

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- $(a^m)^n = a^{mn}$

where $a > 0$, $b > 0$ and $m, n \in \mathbb{R}$

VISIT:

The following two videos explain the exponent laws.

Part 1:

▶ See video: [2DZD](#) at www.everythingmaths.co.za

Part 2:

▶ See video: [2DZF](#) at www.everythingmaths.co.za

Worked example 1: Applying the exponential laws

QUESTION

Simplify:

1. $2^{3x} \times 2^{4x}$
2. $\frac{4x^3}{2x^5}$
3. $\frac{12p^2t^5}{3pt^3}$
4. $(3x)^2$
5. $(3^45^2)^3$
6. $6p^0 \times (7p)^0$
7. $\left(\frac{2xp}{6x^2}\right)^3$
8. $(2^{-2})^{2x+1}$

SOLUTION

1. $2^{3x} \times 2^{4x} = 2^{3x+4x} = 2^{7x}$
2. $\frac{4x^3}{2x^5} = 2x^{3-5} = 2x^{-2} = \frac{2}{x^2}$
3. $\frac{12p^2t^5}{3pt^3} = 4p^{(2-1)}t^{(5-3)} = 4pt^2$
4. $(3x)^2 = 3^2x^2 = 9x^2$
5. $(3^4 \times 5^2)^3 = 3^{(4 \times 3)} \times 5^{(2 \times 3)} = 3^{12} \times 5^6$
6. $6p^0 \times (7p)^0 = 6(1) \times 1 = 6$
7. $\left(\frac{2xp}{6x^2}\right)^3 = \left(\frac{p}{3x}\right)^3 = \frac{p^3}{27x^3}$
8. $(2^{-2})^{2x+1} = 2^{-2(2x+1)} = 2^{-4x-2}$

NOTE:

When you have a fraction that is one term over one term, use the method of Finding Prime Bases - in other words use prime factorisation on the bases.

Worked example 2: Exponential expressions

QUESTION

Simplify: $\frac{2^{2n} \times 4^n \times 2}{16^n}$

SOLUTION

Step 1: Change the bases to prime numbers

At first glance it appears that we cannot simplify this expression. However, if we reduce the bases to prime bases, then we can apply the exponent laws.

$$\frac{2^{2n} \times 4^n \times 2}{16^n} = \frac{2^{2n} \times (2^2)^n \times 2^1}{(2^4)^n}$$

Step 2: Simplify the exponents

$$\begin{aligned} &= \frac{2^{2n} \times 2^{2n} \times 2^1}{2^{4n}} \\ &= \frac{2^{2n+2n+1}}{2^{4n}} \\ &= \frac{2^{4n+1}}{2^{4n}} \\ &= 2^{4n+1-(4n)} \\ &= 2 \end{aligned}$$

Worked example 3: Exponential expressions

QUESTION

Simplify:

$$\frac{5^{2x-1} \cdot 9^{x-2}}{15^{2x-3}}$$

SOLUTION

Step 1: Change the bases to prime numbers

$$\begin{aligned} \frac{5^{2x-1} \cdot 9^{x-2}}{15^{2x-3}} &= \frac{5^{2x-1} \cdot (3^2)^{x-2}}{(5 \times 3)^{2x-3}} \\ &= \frac{5^{2x-1} \cdot 3^{2x-4}}{5^{2x-3} \cdot 3^{2x-3}} \end{aligned}$$

Step 2: Subtract the exponents (same base)

$$\begin{aligned} &= 5^{(2x-1)-(2x-3)} \times 3^{(2x-4)-(2x-3)} \\ &= 5^{2x-1-2x+3} \times 3^{2x-4-2x+3} \\ &= 5^2 \times 3^{-1} \end{aligned}$$

Step 3: Write the answer as a fraction

$$\begin{aligned} &= \frac{5^2}{3} \\ &= \frac{25}{3} \end{aligned}$$

NOTE:

When working with exponents, all the laws of operation for algebra apply.

Worked example 4: Simplifying by taking out a common factor**QUESTION**

Simplify:

$$\frac{2^t - 2^{t-2}}{3 \cdot 2^t - 2^t}$$

SOLUTION**Step 1: Simplify to a form that can be factorised**

For each of the exponent laws we can “undo” the law - in other words we can work backwards. For this expression we can reverse the multiplication law to write 2^{t-2} as $2^t \cdot 2^{-2}$.

$$\frac{2^t - 2^{t-2}}{3 \cdot 2^t - 2^t} = \frac{2^t - (2^t \cdot 2^{-2})}{3 \cdot 2^t - 2^t}$$

Step 2: Take out a common factor

$$= \frac{2^t (1 - 2^{-2})}{2^t (3 - 1)}$$

Step 3: Cancel the common factor and simplify

$$\begin{aligned} &= \frac{1 - 2^{-2}}{3 - 1} \\ &= \frac{1 - \frac{1}{4}}{2} \\ &= \frac{3}{4} \times \frac{1}{2} \\ &= \frac{3}{8} \end{aligned}$$

NOTE:

When you have a fraction that has more than one term in the numerator or denominator, change to prime bases if necessary and then factorise.

Worked example 5: Simplifying using difference of two squares**QUESTION**

Simplify:

$$\frac{9^x - 1}{3^x + 1}$$

SOLUTION

Step 1: Change the bases to prime numbers

$$\begin{aligned}\frac{9^x - 1}{3^x + 1} &= \frac{(3^2)^x - 1}{3^x + 1} \\ &= \frac{3^{2x} - 1}{3^x + 1} \quad \text{Recognise that } 3^{2x} = (3^x)^2\end{aligned}$$

Step 2: Factorise using the difference of squares

$$= \frac{(3^x - 1)(3^x + 1)}{3^x + 1}$$

Step 3: Cancel the common factor and simplify

$$= 3^x - 1$$

Exercise 2 – 1:

Simplify without using a calculator:

1. 16^0

2. $16a^0$

3. $11^{9x} \times 11^{2x}$

4. $10^{6x} \times 10^{2x}$

5. $(6c)^3$

6. $(5n)^3$

7. $\frac{2^{-2}}{3^2}$

8. $\frac{5}{2^{-3}}$

9. $\left(\frac{2}{3}\right)^{-3}$

10. $\frac{a^2}{a^{-1}}$

11. $\frac{xy^{-3}}{x^4y}$

12. x^2x^{3t+1}

13. $3 \times 3^{2a} \times 3^2$

14. $\frac{2^{m+20}}{2^{m+20}}$

15. $\frac{2^{x+4}}{2^{x+3}}$

16. $(2a^4)(3ab^2)$

17. $(7m^4n)(8m^6n^8)$

18. $2(-a^7b^8)(-4a^3b^6)(-9a^6b^2)$

19. $(-9x^3y^6)\left(\frac{1}{9}x^8y^7\right)\left(\frac{1}{5}x^3y^6\right)$

20. $\frac{a^{3x}}{a^x}$

21. $\frac{20x^{10}a^4}{4x^9a^3}$

22. $\frac{18c^{10}p^8}{9c^6p^5}$

23. $\frac{6m^8a^{10}}{2m^3a^5}$

24. $3^{12} \div 3^9$

25. $\frac{7(a^3)^3}{a^7}$

26. $\frac{9(ab^4)^8}{a^3b^5}$

27. $\frac{2^2}{6^2}$

28. $\left(\frac{a^6}{b^7}\right)^5$

29. $(2t^4)^3$

30. $(3^{n+3})^2$

31. $\frac{3^n9^{n-3}}{27^{n-1}}$

32. $\frac{13^c + 13^{c+2}}{3 \times 13^c - 13^c}$

33. $\frac{3^{5x} \times 81^{5x} \times 3^3}{9^{8x}}$

$$34. \frac{16^x - 144^b}{4^x - 12^b}$$

$$37. \frac{9^3 \times 20^2}{4 \times 5^2 \times 3^5}$$

$$35. \frac{5^{2y-3} 2^{4y+4}}{10^{-5y+5}}$$

$$38. \frac{7^b + 7^{b-2}}{4 \times 7^b + 3 \times 7^b}$$

$$36. \frac{6^4 \times 12^3 \times 4^5}{30^3 \times 3^6}$$

$$39. \frac{12^y - 96^y}{3^y + 6^y}$$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2DZG | 2. 2DZH | 3. 2DZJ | 4. 2DZK | 5. 2DZM | 6. 2DZN | 7. 2DZP | 8. 2DZQ |
| 9. 2DZR | 10. 2DZS | 11. 2DZT | 12. 2DZV | 13. 2DZW | 14. 2DZX | 15. 2DZY | 16. 2DZZ |
| 17. 2F22 | 18. 2F23 | 19. 2F24 | 20. 2F25 | 21. 2F26 | 22. 2F27 | 23. 2F28 | 24. 2F29 |
| 25. 2F2B | 26. 2F2C | 27. 2F2D | 28. 2F2F | 29. 2F2G | 30. 2F2H | 31. 2F2J | 32. 2F2K |
| 33. 2F2M | 34. 2F2N | 35. 2F2P | 36. 2F2Q | 37. 2F2R | 38. 2F2S | 39. 2F2T | |



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2.3 Rational exponents

EMAV

We can also apply the exponent laws to expressions with rational exponents.

Worked example 6: Simplifying rational exponents

QUESTION

Simplify:

$$2x^{\frac{1}{2}} \times 4x^{-\frac{1}{2}}$$

SOLUTION

$$\begin{aligned}
 2x^{\frac{1}{2}} \times 4x^{-\frac{1}{2}} &= 8x^{\frac{1}{2} - \frac{1}{2}} \\
 &= 8x^0 \\
 &= 8(1) \\
 &= 8
 \end{aligned}$$

Worked example 7: Simplifying rational exponents

QUESTION

Simplify:

$$(0,008)^{\frac{1}{3}}$$

SOLUTION

Step 1: Write as a fraction and simplify

$$\begin{aligned}(0,008)^{\frac{1}{3}} &= \left(\frac{8}{1000}\right)^{\frac{1}{3}} \\&= \left(\frac{1}{125}\right)^{\frac{1}{3}} \\&= \left(\frac{1}{5^3}\right)^{\frac{1}{3}} \\&= \frac{1^{\frac{1}{3}}}{5^{(3 \cdot \frac{1}{3})}} \\&= \frac{1}{5}\end{aligned}$$

VISIT:

Extension: the following video provides a summary of all the exponent rules and rational exponents.

▶ See video: 2F2V at www.everythingmaths.co.za

Exercise 2 – 2:

Simplify without using a calculator:

1. $t^{\frac{1}{4}} \times 3t^{\frac{7}{4}}$
2. $\frac{16x^2}{(4x^2)^{\frac{1}{2}}}$
3. $(0,25)^{\frac{1}{2}}$
4. $(27)^{-\frac{1}{3}}$
5. $(3p^2)^{\frac{1}{2}} \times (3p^4)^{\frac{1}{2}}$
6. $12(a^4b^8)^{\frac{1}{2}} \times (512a^3b^3)^{\frac{1}{3}}$
7. $((-2)^4a^6b^2)^{\frac{1}{2}}$
8. $(a^{-2}b^6)^{\frac{1}{2}}$
9. $(16x^{12}b^6)^{\frac{1}{3}}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2F2W
2. 2F2X
3. 2F2Y
4. 2F2Z
5. 2F32
6. 2F33
7. 2F34
8. 2F35
9. 2F36



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Exponential equations have the unknown variable in the exponent. Here are some examples:

$$3^{x+1} = 9$$
$$5^t + 3 \times 5^{t-1} = 400$$

If we can write a single term with the same base on each side of the equation, we can equate the exponents. This is one method to solve exponential equations.

Important: if $a > 0$ and $a \neq 1$ then:

$$a^x = a^y$$

then $x = y$ (same base)

Also notice that if $a = 1$, then x and y can be different.

Worked example 8: Equating exponents

QUESTION

Solve for x : $3^{x+1} = 9$.

SOLUTION

Step 1: Change the bases to prime numbers

$$3^{x+1} = 3^2$$

Step 2: The bases are the same so we can equate exponents

$$x + 1 = 2$$
$$\therefore x = 1$$

Worked example 9: Equating exponents

QUESTION

Solve for t : $3^t = 1$.

SOLUTION

Step 1: Solve for t

We know from the exponent identities that $a^0 = 1$, therefore:

$$3^t = 1$$
$$3^t = 3^0$$
$$\therefore t = 0$$

Worked example 10: Solving equations by taking out a common factor

QUESTION

Solve for t : $5^t + 3 \cdot 5^{t+1} = 400$.

SOLUTION

Step 1: Rewrite the expression

$$5^t + 3(5^t \cdot 5) = 400$$

Step 2: Take out a common factor

$$5^t(1 + 3 \cdot 5) = 400$$

$$5^t(1 + 15) = 400$$

Step 3: Simplify

$$5^t(16) = 400$$

$$5^t = 25$$

Step 4: Change the bases to prime numbers

$$5^t = 5^2$$

Step 5: The bases are the same so we can equate exponents

$$\therefore t = 2$$

Worked example 11: Solving equations by factorising a trinomial

QUESTION

Solve for x :

$$3^{2x} - 80 \cdot 3^x - 81 = 0$$

SOLUTION

Step 1: Factorise the trinomial

$$(3^x - 81)(3^x + 1) = 0$$

Step 2: Solve for x

$3^x = 81$ or $3^x = -1$. However $3^x = -1$ is undefined, so:

$$3^x = 81$$

$$3^x = 3^4$$

$$x = 4$$

Therefore $x = 4$

Worked example 12: Solving equations by factorising a trinomial**QUESTION**

Solve for p :

$$p - 13p^{\frac{1}{2}} + 36 = 0$$

SOLUTION**Step 1: Rewrite the equation**

We notice that $\left(p^{\frac{1}{2}}\right)^2 = p$ so we can rewrite the equation as:

$$\left(p^{\frac{1}{2}}\right)^2 - 13p^{\frac{1}{2}} + 36 = 0$$

Step 2: Factorise as a trinomial

$$\left(p^{\frac{1}{2}} - 9\right)\left(p^{\frac{1}{2}} - 4\right) = 0$$

Step 3: Solve to find both roots

$$p^{\frac{1}{2}} - 9 = 0 \quad \text{or} \quad p^{\frac{1}{2}} - 4 = 0$$

$$p^{\frac{1}{2}} = 9 \quad \quad \quad p^{\frac{1}{2}} = 4$$

$$\left(p^{\frac{1}{2}}\right)^2 = (9)^2 \quad \quad \quad \left(p^{\frac{1}{2}}\right)^2 = (4)^2$$

$$p = 81 \quad \quad \quad p = 16$$

Therefore $p = 81$ or $p = 16$.

Worked example 13: Solving equations by factorisation**QUESTION**

Solve for x :

$$2^x - 2^{4-x} = 0$$

SOLUTION

Step 1: Rewrite the equation

In order to get the equation into a form which we can factorise, we need to rewrite the equation:

$$\begin{aligned}2^x - 2^{4-x} &= 0 \\2^x - 2^4 \cdot 2^{-x} &= 0 \\2^x - \frac{2^4}{2^x} &= 0\end{aligned}$$

Now eliminate the fraction by multiplying both sides of the equation by the denominator, 2^x .

$$\begin{aligned}\left(2^x - \frac{2^4}{2^x}\right) \times 2^x &= 0 \times 2^x \\2^{2x} - 16 &= 0\end{aligned}$$

Step 2: Factorise the equation

Now that we have rearranged the equation, we can see that we are left with a difference of two squares. Therefore:

$$\begin{aligned}2^{2x} - 16 &= 0 \\(2^x - 4)(2^x + 4) &= 0 \\2^x &= 4 \quad 2^x \neq -4 \quad (\text{a positive integer with an exponent is always positive}) \\2^x &= 2^2 = 4 \\x &= 2\end{aligned}$$

Therefore $x = 2$.

Exercise 2 – 3:

1. Solve for the variable:

a) $2^{x+5} = 32$

c) $64^{y+1} = 16^{2y+5}$

e) $25 = 5^{z-4}$

g) $81^{k+2} = 27^{k+4}$

i) $27^x \times 9^{x-2} = 1$

k) $(7^x - 49)(3^x - 27) = 0$

m) $(10^x - 1)(3^x - 81) = 0$

o) $9^m + 3^{3-2m} = 28$

q) $4^{x+3} = 0,5$

b) $5^{2x+2} = \frac{1}{125}$

d) $3^{9x-2} = 27$

f) $-\frac{1}{2} \cdot 6^{\frac{m}{2}+3} = -18$

h) $25^{1-2x} - 5^4 = 0$

j) $2^t + 2^{t+2} = 40$

l) $(2 \cdot 2^x - 16)(3^{x+1} - 9) = 0$

n) $2 \times 5^{2-x} = 5 + 5^x$

p) $y - 2y^{\frac{1}{2}} + 1 = 0$

r) $2^a = 0,125$

$$s) 10^x = 0,001$$

$$u) \frac{8^x - 1}{2^x - 1} = 8 \cdot 2^x + 9$$

$$t) 2^{x^2 - 2x - 3} = 1$$

$$v) \frac{27^x - 1}{9^x + 3^x + 1} = -\frac{8}{9}$$

2. The growth of algae can be modelled by the function $f(t) = 2^t$. Find the value of t such that $f(t) = 128$.
3. Use trial and error to find the value of x correct to 2 decimal places
 $2^x = 7$
4. Use trial and error to find the value of x correct to 2 decimal places
 $5^x = 11$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 2F37 | 1b. 2F38 | 1c. 2F39 | 1d. 2F3B | 1e. 2F3C | 1f. 2F3D |
| 1g. 2F3F | 1h. 2F3G | 1i. 2F3H | 1j. 2F3J | 1k. 2F3K | 1l. 2F3M |
| 1m. 2F3N | 1n. 2F3P | 1o. 2F3Q | 1p. 2F3R | 1q. 2F3S | 1r. 2F3T |
| 1s. 2F3V | 1t. 2F3W | 1u. 2F3X | 1v. 2F3Y | 2. 2F3Z | 3. 2F42 |
| 4. 2F43 | | | | | |



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2.5 Summary

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📺 See presentation: [2F44](#) at www.everythingmaths.co.za

- Exponential notation means writing a number as a^n where n is any natural number and a is any real number.
- a is the base and n is the exponent or index.
- Definition:
 - $a^n = a \times a \times \cdots \times a$ (n times)
 - $a^0 = 1$, if $a \neq 0$
 - $a^{-n} = \frac{1}{a^n}$, if $a \neq 0$
 - $\frac{1}{a^{-n}} = a^n$, if $a \neq 0$
- The laws of exponents:
 - $a^m \times a^n = a^{m+n}$
 - $\frac{a^m}{a^n} = a^{m-n}$
 - $(ab)^n = a^n b^n$
 - $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
 - $(a^m)^n = a^{mn}$
- When simplifying expressions with exponents, we can reduce the bases to prime bases or factorise.
- When solving equations with exponents, we can apply the rule that if $a^x = a^y$ then $x = y$; or we can factorise the expressions.

1. Simplify:

a) $(8x)^3$

d) $15^{3x} \times 15^{12x}$

g) $(\frac{1}{7}a^2b^9)(6a^6b^2)(-3a^7b)$

j) $\frac{2(x^4)^3}{x^{12}}$

m) $\frac{6^{5p}}{9^p}$

p) $\frac{5^{b-3}}{5^{b+1}}$

s) $\frac{3^3}{9^3}$

v) $\left(\frac{2x^{2a}}{y^{-b}}\right)^3$

y) $\frac{(-3)^{-3}(-3)^2}{(-3)^{-4}}$

b) $t^3 \times 2t^0$

e) $\frac{7^{y+7}}{7^{y+6}}$

h) $(b^{k+1})^k$

k) $\frac{a^6b^5}{7(a^8b^3)^2}$

n) $m^{-2t} \times (3m^t)^3$

q) $\frac{2^{a-2}3^{a+3}}{6^a}$

t) $\frac{x^{-1}}{x^4y^{-2}}$

w) $\frac{2^{3x-1}8^{x+1}}{4^{2x-2}}$

z) $(3^{-1} + 2^{-1})^{-1}$

c) $5^{2x+y} \times 5^{3(x+z)}$

f) $3(d^4)(7d^3)$

i) $\frac{24c^8m^7}{6c^2m^5}$

l) $\left(\frac{a^7}{b^4}\right)^2$

o) $\frac{3x^{-3}}{(3x)^2}$

r) $\frac{3^n9^{n-3}}{27^{n-1}}$

u) $\frac{(-1)^4}{(-2)^{-3}}$

x) $\frac{6^{2x}11^{2x}}{22^{2x-1}3^{2x}}$

2. Simplify:

a) $\frac{9^{n-1}27^{3-2n}}{81^{2-n}}$

d) $\frac{2^{3p}+1}{2^p+1}$

g) $\frac{13^a+13^{a+2}}{6 \times 13^a - 13^a}$

j) $\frac{11^{-4c-4}4^{4c-3}}{22^{-6c-2}}$

m) $(0,81)^{\frac{1}{2}}$

p) $\frac{a^{-1}-b^{-1}}{a-b}$

s) $(a^{\frac{1}{2}} + a^{-\frac{1}{2}})^2 - (a^{\frac{1}{2}} - a^{-\frac{1}{2}})^2$

b) $\frac{2^{3n+2} \cdot 8^{n-3}}{4^{3n-2}}$

e) $(a^{10}b^6)^{\frac{1}{2}}$

h) $\frac{3^{8z} \times 27^{8z} \times 3^2}{9^{6z}}$

k) $\frac{12^4 \times 2^4}{16^6 \times 10}$

n) $12(a^{10}b^{20})^{\frac{1}{5}} \times (729a^{12}b^{15})^{\frac{1}{3}}$

q) $\left((x^{36})^{\frac{1}{2}}\right)^{\frac{1}{3}}$

c) $\frac{3^{t+3} + 3^t}{2 \times 3^t}$

f) $(9x^8y^4)^{\frac{1}{2}}$

i) $\frac{121^b - 16^p}{11^b + 4^p}$

l) $\frac{5^6 \times 3^{16} \times 2^7}{10^8 \times 9^6}$

o) $2(p^{30}q^{20})^{\frac{1}{5}} \times (1331p^{12}q^6)^{\frac{1}{3}}$

r) $\left(\frac{2}{3}\right)^{x+y} \cdot \left(\frac{3}{2}\right)^{x-y}$

3. Solve:

a) $3^x = \frac{1}{27}$

c) $5^{t-1} = 1$

e) $-\frac{64}{3} = -\frac{4}{3}2^{-\frac{c}{3}+1}$

g) $2^{m+1} = (0,5)^{m-2}$

i) $z^{\frac{3}{2}} = 64$

b) $121 = 11^{m-1}$

d) $2 \times 7^{3x} = 98$

f) $-\frac{1}{2}6^{-n-3} = -18$

h) $3^{y+1} = 5^{y+1}$

j) $16x^{\frac{1}{2}} - 4 = 0$

$$k) m^0 + m^{-1} = 0$$

$$m) 3^p + 3^p + 3^p = 27$$

$$o) x^{\frac{1}{2}} + 3x^{\frac{1}{4}} - 18 = 0$$

$$q) (2^x - 8)(3^x - 9) = 0$$

$$s) 5 \cdot 2^{x^2+1} = 20$$

$$u) \frac{8^x - 1}{2^x - 1} = 7$$

$$w) \frac{a^{3x} \cdot a^{\frac{1}{x}}}{a^{-4}} = 1$$

$$l) t^{\frac{1}{2}} - 3t^{\frac{1}{4}} + 2 = 0$$

$$n) k^{-1} - 7k^{-\frac{1}{2}} - 18 = 0$$

$$p) \frac{16^x - 1}{4^2x + 1} = 3$$

$$r) (6^x - 36)(16 - 4^x) = 0$$

$$t) 27^{x-2} = 9^{2x+1}$$

$$v) \frac{35^x}{5^x} = \frac{1}{7}$$

$$x) 2x^{\frac{1}{2}} + 1 = -x$$

4. Use trial and error to find the value of x correct to 2 decimal places

$$4^x = 44$$

5. Use trial and error to find the value of x correct to 2 decimal places

$$3^x = 30$$

6. Explain why the following statements are false:

$$a) \frac{1}{a^{-1} + b^{-1}} = a + b$$

$$b) (a + b)^2 = a^2 + b^2$$

$$c) \left(\frac{1}{a^2}\right)^{\frac{1}{3}} = a^{\frac{2}{3}}$$

$$d) 2 \cdot 3^x = 6^x$$

$$e) x^{-\frac{1}{2}} = \frac{1}{-x^{\frac{1}{2}}}$$

$$f) (3x^4y^2)^3 = 3x^{12}y^6$$

7. If $2^{2013} \cdot 5^{2015}$ is written out in full how many digits will there be?

8. Prove that $\frac{2^{n+1} + 2^n}{2^n - 2^{n-1}} = \frac{3^{n+1} + 3^n}{3^n - 3^{n-1}}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2F45	1b. 2F46	1c. 2F47	1d. 2F48	1e. 2F49	1f. 2F4B
1g. 2F4C	1h. 2F4D	1i. 2F4F	1j. 2F4G	1k. 2F4H	1l. 2F4J
1m. 2F4K	1n. 2F4M	1o. 2F4N	1p. 2F4P	1q. 2F4Q	1r. 2F4R
1s. 2F4S	1t. 2F4T	1u. 2F4V	1v. 2F4W	1w. 2F4X	1x. 2F4Y
1y. 2F4Z	1z. 2F52	2a. 2F53	2b. 2F54	2c. 2F55	2d. 2F56
2e. 2F57	2f. 2F58	2g. 2F59	2h. 2F5B	2i. 2F5C	2j. 2F5D
2k. 2F5F	2l. 2F5G	2m. 2F5H	2n. 2F5J	2o. 2F5K	2p. 2F5M
2q. 2F5N	2r. 2F5P	2s. 2F5Q	3a. 2F5R	3b. 2F5S	3c. 2F5T
3d. 2F5V	3e. 2F5W	3f. 2F5X	3g. 2F5Y	3h. 2F5Z	3i. 2F62
3j. 2F63	3k. 2F64	3l. 2F65	3m. 2F66	3n. 2F67	3o. 2F68
3p. 2F69	3q. 2F6B	3r. 2F6C	3s. 2F6D	3t. 2F6F	3u. 2F6G
3v. 2F6H	3w. 2F6J	3x. 2F6K	4. 2F6M	5. 2F6N	6a. 2F6P
6b. 2F6Q	6c. 2F6R	6d. 2F6S	6e. 2F6T	6f. 2F6V	7. 2F6W
8. 2F6X					



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Number patterns

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3.2	<i>Describing sequences</i>	60
3.3	<i>Chapter summary</i>	68

3.1 Introduction

EMAX

In earlier grades you saw patterns in the form of pictures and numbers. In this chapter, we learn more about the mathematics of patterns. Patterns are repetitive sequences and can be found in nature, shapes, events, sets of numbers and almost everywhere you care to look. For example, seeds in a sunflower, snowflakes, geometric designs on quilts or tiles, or the number sequence $0; 4; 8; 12; 16; \dots$



Figure 3.1: The pattern of seeds within a sunflower follows the Fibonacci sequence, or $1; 2; 3; 5; 8; 13; 21; 34; 55; 89; 144; \dots$

VISIT:

Interested in learning more about the relationship between Fibonacci sequences and sunflowers?

► See video: [2F6Y](https://www.everythingmaths.co.za/2F6Y) at www.everythingmaths.co.za

Try spot any patterns in the following sequences on your own:

1. $2; 4; 6; 8; 10; \dots$
2. $1; 2; 4; 7; 11; \dots$
3. $1; 4; 9; 16; 25; \dots$
4. $5; 10; 20; 40; 80; \dots$

VISIT:

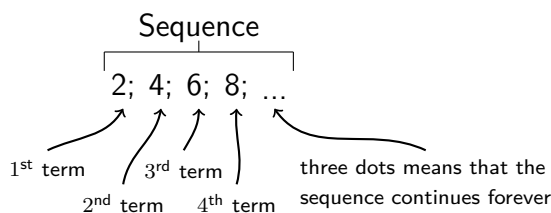
Identifying patterns in sequences.

► See video: [2F6Z](https://www.everythingmaths.co.za/2F6Z) at www.everythingmaths.co.za

3.2 Describing sequences

EMAY

A sequence is an ordered list of items, usually numbers. Each item which makes up a sequence is called a “term”.



Sequences can have interesting patterns. Here we examine some types of patterns and how they are formed.

Examples:

1. 1; 4; 7; 10; 13; 16; 19; 22; 25; ...

There is difference of 3 between successive terms.

The pattern is continued by adding 3 to the previous term.

2. 13; 8; 3; -2; -7; -12; -17; -22; ...

There is a difference of -5 between successive terms.

The pattern is continued by adding -5 to (i.e. subtracting 5 from) the previous term.

3. 2; 4; 8; 16; 32; 64; 128; 256; ...

This sequence has a factor of 2 between successive terms.

The pattern is continued by multiplying the previous term by 2.

4. 3; -9; 27; -81; 243; -729; 2187; ...

This sequence has a factor of -3 between successive terms.

The pattern is continued by multiplying the previous term by -3.

5. 9; 3; 1; $\frac{1}{3}$; $\frac{1}{9}$; $\frac{1}{27}$; ...

This sequence has a factor of $\frac{1}{3}$ between successive terms.

The pattern is continued by multiplying the previous term by $\frac{1}{3}$ which is equivalent to dividing the previous term by 3.

Worked example 1: Study table

QUESTION

You and 3 friends decide to study for Maths and are sitting together at a square table. A few minutes later, 2 other friends arrive and would like to sit at your table. You move another table next to yours so that 6 people can sit at the table. Another 2 friends also want to join your group, so you take a third table and add it to the existing tables. Now 8 people can sit together.

Examine how the number of people sitting is related to the number of tables. Is there a pattern?

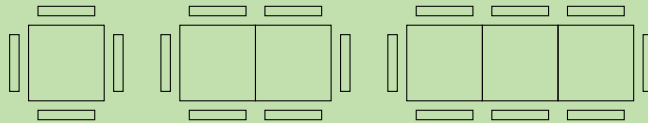


Figure 3.2: Two more people can be seated for each table added.

SOLUTION

Step 1: Make a table to see if a pattern forms

Number of tables, n	Number of people seated
1	$4 = 4$
2	$4 + 2 = 6$
3	$4 + 2 + 2 = 8$
4	$4 + 2 + 2 + 2 = 10$
$\vdots$	$\vdots$
n	$4 + 2 + 2 + 2 + \dots + 2$

Step 2: Describe the pattern

We can see that for 3 tables we can seat 8 people, for 4 tables we can seat 10 people and so on. We started out with 4 people and added two each time. So for each table added, the number of people increased by 2.

So the pattern formed is 4; 6; 8; 10; ...

To describe terms in a number pattern we use the following notation:

The first term of a sequence is T_1 .

The fourth term of a sequence is T_4 .

The tenth term of a sequence is T_{10} .

The general term is often expressed as the n^{th} term and is written as T_n .

A sequence does not have to follow a pattern but, when it does, we can write down the general formula to calculate any term. For example, consider the following linear sequence: 1; 3; 5; 7; 9; ...

The n^{th} term is given by the general formula: $T_n = 2n - 1$

You can check this by substituting values into the formula:

$$\begin{aligned}T_1 &= 2(1) - 1 = 1 \\T_2 &= 2(2) - 1 = 3 \\T_3 &= 2(3) - 1 = 5 \\T_4 &= 2(4) - 1 = 7 \\T_5 &= 2(5) - 1 = 9\end{aligned}$$

If we find the relationship between the position of a term and its value, we can find a general formula which matches the pattern and find any term in the sequence.

Common difference

EMAZ

Consider the following sequence:

$$6; 1; -4; -9; \dots$$

We can see that each term is decreasing by 5 but how would we determine the general formula for the n^{th} term? Let us try to do this with a table.

Term number	T_1	T_2	T_3	T_4	T_n
Term	6	1	-4	-9	T_n
Formula	$6 - 0 \times 5$	$6 - 1 \times 5$	$6 - 2 \times 5$	$6 - 3 \times 5$	$6 - (n - 1) \times 5$

You can see that the difference between the successive terms is always the coefficient of n in the formula. This is called a **common difference**.

Therefore, for sequences with a common difference, the general formula will always be of the form: $T_n = dn + c$ where d is the difference between each term and c is some constant.

NOTE:

Sequences with a common difference are called linear sequences.

DEFINITION: Common difference

The common difference is the difference between any term and the term before it. The common difference is denoted by d .

For example, consider the sequence 10; 7; 4; 1; ...

To calculate the common difference, we find the difference between any term and the previous term.

Let us find the common difference between the first two terms.

$$\begin{aligned} d &= T_2 - T_1 \\ &= 7 - 10 \\ &= -3 \end{aligned}$$

Let us check another two terms:

$$\begin{aligned} d &= T_4 - T_3 \\ &= 1 - 4 \\ &= -3 \end{aligned}$$

We see that d is constant.

In general: $d = T_n - T_{n-1}$

IMPORTANT!

$d \neq T_{n-1} - T_n$ for example, $d = T_2 - T_1$, not $T_1 - T_2$.

Worked example 2: Study table, continued

QUESTION

As before, you and 3 friends are studying for Maths and are sitting together at a square table. A few minutes later 2 other friends arrive so you move another table next to yours. Now 6 people can sit at the table. Another 2 friends also join your group, so you take a third table and add it to the existing tables. Now 8 people can sit together as shown below.

1. Find an expression for the number of people seated at n tables.
2. Use the general formula to determine how many people can sit around 12 tables.
3. How many tables are needed to seat 20 people?

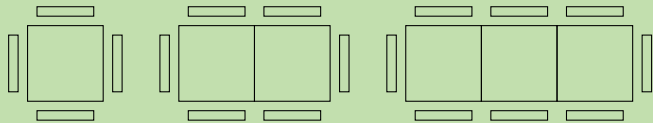


Figure 3.3: Two more people can be seated for each table added.

SOLUTION

Step 1: Make a table to see the pattern

Number of Tables, n	Number of people seated	Pattern
1	$4 = 4$	$= 4 + 2(0)$
2	$4 + 2 = 6$	$= 4 + 2(1)$
3	$4 + 2 + 2 = 8$	$= 4 + 2(2)$
4	$4 + 2 + 2 + 2 = 10$	$= 4 + 2(3)$
$\vdots$	$\vdots$	$\vdots$
n	$4 + 2 + 2 + 2 + \dots + 2$	$= 4 + 2(n - 1)$

Note: There may be variations in how you think of the pattern in this problem. For example, you may view this problem as the person on one end fixed, two people seated opposite each other per table and one person at the other end fixed. This results in $1 + 2n + 1 = 2n + 2$. Your formula for T_n will still be correct.

Step 2: Describe the pattern

The number of people seated at n tables is $T_n = 4 + 2(n - 1)$

Step 3: Calculate the 12th term, in other words, find T_n if $n = 12$

$$\begin{aligned}
 T_{12} &= 4 + 2(12 - 1) \\
 &= 4 + 2(11) \\
 &= 4 + 22 \\
 &= 26
 \end{aligned}$$

Therefore 26 people can be seated at 12 tables.

Step 4: Calculate the number of tables needed to seat 20 people, in other words, find n if $T_n = 20$

$$\begin{aligned}
 T_n &= 4 + 2(n - 1) \\
 20 &= 4 + 2(n - 1) \\
 20 &= 4 + 2n - 2 \\
 20 - 4 + 2 &= 2n \\
 18 &= 2n \\
 \frac{18}{2} &= n \\
 n &= 9
 \end{aligned}$$

Therefore 9 tables are needed to seat 20 people.

It is important to note the difference between n and T_n . n can be compared to a place holder indicating the position of the term in the sequence, while T_n is the value of the place held by n . From our example above, the first table holds 4 people. So for $n = 1$, the value of $T_1 = 4$ and so on:

n	1	2	3	4	...
T_n	4	6	8	10	...

Worked example 3: Data plans**QUESTION**

Raymond subscribes to a limited data plan from Vodacell. The limited data plans cost R 120 for 1 gigabyte (GB) per month, R 135 for 2 GB per month and R 150 for 3 GB per month. Assume this pattern continues indefinitely.

1. Use a table to set up the pattern of the cost of the data plans.
2. Find the general formula for the sequence.
3. Use the general formula to determine the cost for a 30 GB data plan.
4. The cost of an unlimited data plan is R 520 per month. Determine the amount of data Raymond would have to use for it to be cheaper for him to sign up for the unlimited plan.

SOLUTION

Step 1: Make a table to see the pattern

Number of GB (n)	1	2	3	4
Cost (in Rands)	120	135	150	165
Pattern	120	$120 + (1)(15) = 135$	$120 + (2)(15) = 150$	$120 + (3)(15) = 165$

Step 2: Use the observed pattern to determine the general formula.

The price of n GB of data is $T_n = 120 + 15(n - 1)$

Step 3: Determine the cost of 30 GB of data.

This question requires us to determine the value of the 30th term, in other words, find T_n if $n = 30$. Using the general formula, we get:

$$\begin{aligned}T_n &= 120 + 15(n - 1) \\ \therefore T_{30} &= 120 + 15(30 - 1) \\ &= 120 + 15(29) \\ &= 120 + 435 \\ &= 555\end{aligned}$$

Therefore the cost of a 30 GB data package is R 555.

Step 4: Determine when it is cheaper to purchase the unlimited data plan

The final question of this worked example requires us to determine when it would be cheaper for Raymond to purchase an unlimited data plan instead of a limited plan. In other words, we need to find n where T_n is less than R 520.

We know that:

$$T_n = 120 + 15(n - 1)$$

Therefore, if $T_n = 520 = 120 + 15(n - 1)$

Solving for n , we get:

$$\begin{aligned}520 &= 120 + 15(n - 1) \\ 520 &= 120 + 15n - 15 \\ 520 &= 105 + 15n \\ 405 &= 15n \\ \frac{405}{15} &= n \\ n &= 27\end{aligned}$$

Therefore it is cheaper for Raymond to purchase the unlimited data plan if he uses more than 27 GB per month.

VISIT:

Learn more about number patterns.

▶ See video: 2F72 at www.everythingmaths.co.za

Exercise 3 – 1:

1. Use the given pattern to complete the table below.

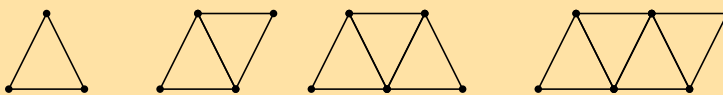
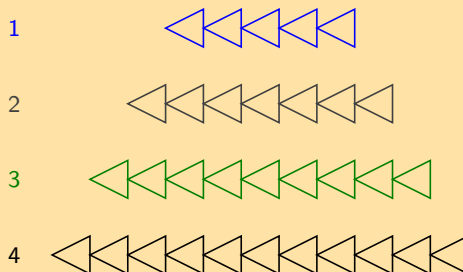


Figure number	1	2	3	4	n
Number of dots					
Number of lines					
Total					

2. Consider the sequence shown here: $-4 ; -1 ; 2 ; 5 ; 8 ; 11 ; 14 ; 17 ; \dots$
If $T_n = 2$ what is the value of T_{n-1} ?
3. Consider the sequence shown here: $C ; D ; E ; F ; G ; H ; I ; J ; \dots$
If $T_n = G$ what is the value of T_{n-4} ?
4. For each of the following sequences determine the common difference. If the sequence is not linear, write "no common difference".
- a) $9 ; -7 ; -8 ; -25 ; -34 ; \dots$ b) $5 ; 12 ; 19 ; 26 ; 33 ; \dots$
c) $2,93 ; 1,99 ; 1,14 ; 0,35 ; \dots$ d) $2,53 ; 1,88 ; 1,23 ; 0,58 ; \dots$
5. Write down the next three terms in each of the following sequences:
- a) $5 ; 15 ; 25 ; \dots$ b) $-8 ; -3 ; 2 ; \dots$
c) $30 ; 27 ; 24 ; \dots$ d) $-13,1 ; -18,1 ; -23,1 ; \dots$
e) $-9x ; -19x ; -29x ; \dots$ f) $-15,8 ; 4,2 ; 24,2 ; \dots$
g) $30b ; 34b ; 38b ; \dots$
6. Given a pattern which starts with the numbers: $3 ; 8 ; 13 ; 18 ; \dots$ determine the values of T_6 and T_9 .
7. Given a sequence which starts with the letters: $C ; D ; E ; F ; \dots$ determine the values of T_5 and T_8 .
8. Given a pattern which starts with the numbers: $7 ; 11 ; 15 ; 19 ; \dots$ determine the values of T_5 and T_8 .
9. The general term is given for each sequence below. Calculate the missing terms (each missing term is represented by $\dots$).
- a) $0 ; 3 ; \dots ; 15 ; 24$ $T_n = n^2 - 1$
b) $3 ; 2 ; 1 ; 0 ; \dots ; -2$ $T_n = -n + 4$
c) $-11 ; \dots ; -7 ; \dots ; -3$ $T_n = -13 + 2n$
d) $1 ; 10 ; 19 ; \dots ; 37$ $T_n = 9n - 8$
e) $9 ; \dots ; 21 ; \dots ; 33$ $T_n = 6n + 3$
10. Find the general formula for the following sequences and then find T_{10} , T_{50} and T_{100}
- a) $2 ; 5 ; 8 ; 11 ; 14 ; \dots$ b) $0 ; 4 ; 8 ; 12 ; 16 ; \dots$ c) $2 ; -1 ; -4 ; -7 ; -10 ; \dots$
11. The diagram below shows pictures which follow a pattern.



- a) How many triangles will there be in the 5th picture?

- b) Determine the formula for the n^{th} term.
 c) Use the formula to find how many triangles are in the 25th picture of the diagram.
12. Study the following sequence: 15 ; 23 ; 31 ; 39 ; ...
 a) Write down the next 3 terms.
 b) Find the general formula for the sequence
 c) Find the value of n if T_n is 191.
13. Study the following sequence: -44 ; -14 ; 16 ; 46 ; ...
 a) Write down the next 3 terms.
 b) Find the general formula for the sequence
 c) Find the value of n if T_n is 406.
14. Consider the following list:
 $-z - 5$; $-4z - 5$; $-6z - 2$; $-8z - 5$; $-10z - 5$; ...
 a) Find the common difference for the terms of the list. If the sequence is not linear (if it does not have a common difference), write "no common difference".
 b) If you are now told that $z = -2$, determine the values of T_1 and T_2 .
15. Consider the following pattern:
 $2n + 4$; 1 ; $-2n - 2$; $-4n - 5$; $-6n - 8$; ...
 a) Find the common difference for the terms of the pattern. If the sequence is not linear (if it does not have a common difference), write "no common difference".
 b) If you are now told that $n = -1$, determine the values of T_1 and T_3 .
16. a) If the following terms make a linear sequence:

$$\frac{k}{3} - 1 ; -\frac{5k}{3} + 2 ; -\frac{2k}{3} + 10 ; \dots$$
 Determine the value of k . If the answer is a non-integer, write the answer as a simplified fraction.
 b) Now determine the numeric value of the first three terms. If the answers are not integers, write your answers as fractions.
17. a) If the following terms make a linear sequence:

$$y - \frac{3}{2} ; -y - \frac{7}{2} ; -7y - \frac{15}{2} ; \dots$$
 find y . If the answer is a non-integer, write the answer as a simplified fraction.
 b) Now determine the numeric value of the first three terms. If the answers are not integers, write your answers as fractions.
18. What is the 649th letter of the sequence:
 PATTERNPATTERNPATTERNPATTERNPATTERNPATTERNPATTE.....?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | |
|--------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|
| 1. 2F73 | 2. 2F74 | 3. 2F75 | 4. 2F76 | 5a. 2F77 | 5b. 2F78 |
| 5c. 2F79 | 5d. 2F7B | 5e. 2F7C | 5f. 2F7D | 5g. 2F7F | 6. 2F7G |
| 7. 2F7H | 8. 2F7J | 9a. 2F7K | 9b. 2F7M | 9c. 2F7N | 9d. 2F7P |
| 9e. 2F7Q | 10a. 2F7R | 10b. 2F7S | 10c. 2F7T | 11. 2F7V | 12. 2F7W |
| 13. 2F7X | 14. 2F7Y | 15. 2F7Z | 16. 2F82 | 17. 2F83 | 18. 2F84 |



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▶ See presentation: 2F85 at www.everythingmaths.co.za

- The general term is expressed as the n^{th} term and is written as T_n .
- We define the common difference d of a sequence as the difference between any two successive terms, where $d = T_n - T(n-1)$
- We can work out a general formula for each number pattern and use it to determine any term in the pattern.

End of chapter Exercise 3 – 2:

1. Analyse the diagram and complete the table.

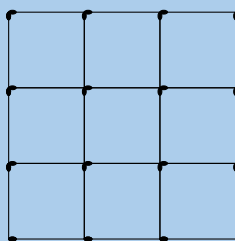


Figure number ($n \times n$)	1×1	2×2	3×3	4×4	$n \times n$
Number of horizontal matches					
Number of vertical matches					
Total number of matches					

2. Given a list of numbers: 7 ; 4 ; 1 ; -2 ; -5 ; ... determine the common difference for the list (if there is one).
3. For the pattern here: -0,55 ; 0,99 ; 2,49 ; 3,91 ; ... calculate the common difference.
If the pattern is not linear, write "no common difference". Otherwise, give your answer as a decimal.
4. Consider the list shown here: 2 ; 7 ; 12 ; 17 ; 22 ; 27 ; 32 ; 37 ; ...
If $T_5 = 22$ what is the value of T_{n-3} ?
5. Write down the next three terms in each of the following linear sequences:
- a) -10,2 ; -29,2 ; -48,2 ; ... b) $50r$; $46r$; $42r$; ...
6. Given a sequence which starts with the numbers: 6 ; 11 ; 16 ; 21 ; ... determine the values of T_6 and T_8 .
7. Given a list which starts with the letters: A ; B ; C ; D ; ... determine the values of T_6 and T_{10} .
8. Find the sixth term in each of the following sequences:
- a) 4 ; 13 ; 22 ; 31 ; ... b) 5 ; 2 ; -1 ; -4 ; ... c) 7,4 ; 9,7 ; 12 ; 14,3 ; ...
9. Find the general formula for the following sequences and then find T_{10} , T_{15} and T_{30}
- a) -18 ; -22 ; -26 ; -30 ; -34 ; ... b) 1 ; -6 ; -13 ; -20 ; -27 ; ...
10. The general term is given for each sequence below. Calculate the missing terms (each missing term is represented by ...).
- a) 10 ; ... ; 14 ; ... ; 18 $T_n = 2n + 8$ b) 2 ; -2 ; -6 ; ... ; -14 $T_n = -4n + 6$
c) 8 ; ... ; 38 ; ... ; 68 $T_n = 15n - 7$

-

-

- Determine the first term.
- Determine the common difference.
- Determine the general formula.
- A row has twenty-five squares. How many matchsticks are there in this row?

- $$-4y - 3; -y; 2y + 3; 5y + 6; 8y + 9; \dots$$

- 69

20. a) If the following terms make a linear sequence:

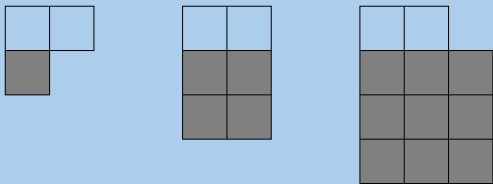
$$2n + \frac{1}{2} ; 3n + \frac{5}{2} ; 7n + \frac{11}{2} ; \dots$$

Determine the value of n . If the answer is a non-integer, write the answer as a simplified fraction.

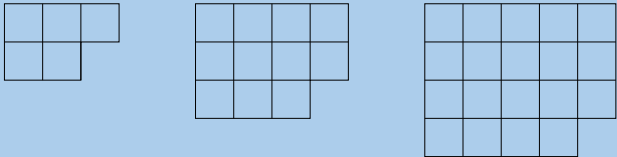
b) Now determine the numeric value of the first three terms. If the answers are not integers, write your answers as fractions.

21. How many blocks will there be in the 85th picture?

Hint: Use the grey blocks to help.

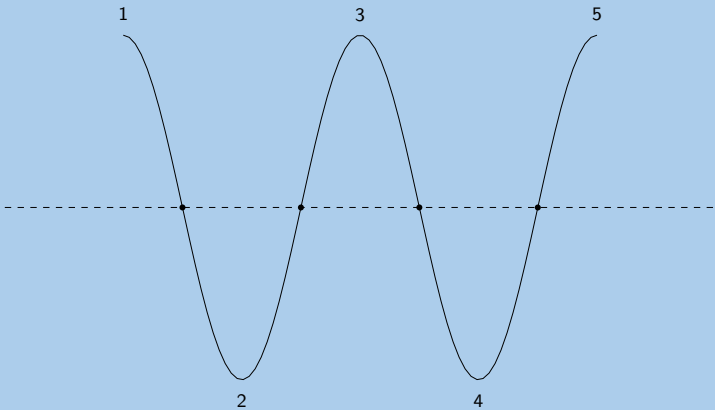


22. Analyse the picture below:



- a) How many blocks are there in the next picture?
- b) Write down the general formula for this pattern.
- c) How many blocks will there be in the 14th picture?

23. A horizontal line intersects a piece of string at 4 points and divides it into five parts, as shown below.



If the piece of string is intersected in this way by 19 parallel lines, each of which intersects it at 4 points, determine the number of parts into which the string will be divided.

24. Use a calculator to explore and then generalise your findings to determine the:

- a) units digit of 3^{2007}
- b) tens digit of 7^{2008}
- c) remainder when 7^{250} is divided by 5

25. Analyse the diagram and complete the table.

The dots follow a triangular pattern and the formula is $T_n = \frac{n(n+1)}{2}$.

The general formula for the lines is $T_n = \frac{3n(n-1)}{2}$.

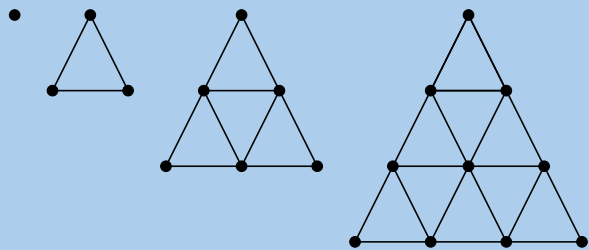


Figure number	1	2	3	4	5	20	n
Number of dots							
Number of lines							
Total							

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1. [2F86](#)
2. [2F88](#)
3. [2F89](#)
4. [2F8B](#)
5. [2F8C](#)
6. [2F8D](#)
7. [2F8F](#)
8a. [2F8G](#)
8b. [2F8H](#)
8c. [2F8J](#)
9. [2F8K](#)
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- 11a. [2F8N](#)
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24a. [2F97](#)
24b. [2F98](#)
- 24c. [2F99](#)
25. [2F87](#)



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Equations and inequalities

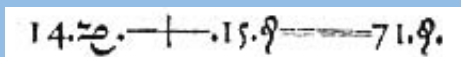
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4.1 Introduction

EMA33

Equations are widely used to describe the world around us. In science equations are used to describe everything from how a ball rolls down a slope to how the planets move around the sun.

In this chapter we will explore different types of equations as well as looking at how these can be used to solve problems in the real world. We will also look at linear inequalities.

DID YOU KNOW?

The first use of an “equals” sign from *The Whetstone of Witte* by Robert Recorde 1557. This equation represents $14x + 15 = 71$. Recorde is also responsible for introducing the pre-existing “plus” sign (+) to the English-speaking world.

4.2 Solving linear equations

EMA34

The simplest equation to solve is a linear equation. A linear equation is an equation where the highest exponent of the variable is 1. The following are examples of linear equations:

$$\begin{aligned} 2x + 2 &= 1 \\ \frac{2 - x}{3x + 1} &= 2 \\ 4(2x - 9) - 4x &= 4 - 6x \\ \frac{2a - 3}{3} - 3a &= \frac{a}{3} \end{aligned}$$

Solving an equation means finding the value of the variable that makes the equation true. For example, to solve the simple equation $x + 1 = 1$, we need to determine the value of x that will make the left hand side equal to the right hand side. The solution is $x = 0$.

The solution, also called the root of an equation, is the value of the variable that satisfies the equation. For linear equations, there is at most one solution for the equation.

To solve equations we use algebraic methods that include expanding expressions, grouping terms, and factorising.

For example:

$$\begin{aligned} 2x + 2 &= 1 \\ 2x &= 1 - 2 \quad (\text{rearrange}) \\ 2x &= -1 \quad (\text{simplify}) \\ x &= -\frac{1}{2} \quad (\text{divide both sides by 2}) \end{aligned}$$

Check the answer by substituting $x = -\frac{1}{2}$.

$$\begin{aligned}\text{LHS} &= 2x + 2 \\ &= 2\left(-\frac{1}{2}\right) + 2 \\ &= -1 + 2 \\ &= 1 \\ \text{RHS} &= 1\end{aligned}$$

Therefore $x = -\frac{1}{2}$

VISIT:

The following video gives an introduction to solving linear equations.

🔊 See video: 2F9B at www.everythingmaths.co.za

Method for solving linear equations

EMA35

The general steps for solving linear equations are:

1. Expand all brackets.
2. Rearrange the terms so that all terms containing the variable are on one side of the equation and all constant terms are on the other side.
3. Group like terms together and simplify.
4. Factorise if necessary.
5. Find the solution and write down the answer.
6. Check the answer by substituting the solution back into the original equation.

IMPORTANT!

An equation must always be balanced, whatever you do to the left-hand side, you must also do to the right-hand side.

Worked example 1: Solving linear equations

QUESTION

Solve for x :

$$4(2x - 9) - 4x = 4 - 6x$$

SOLUTION

Step 1: Expand the brackets and simplify

$$\begin{aligned}4(2x - 9) - 4x &= 4 - 6x \\ 8x - 36 - 4x &= 4 - 6x \\ 8x - 4x + 6x &= 4 + 36 \\ 10x &= 40\end{aligned}$$

Step 2: Divide both sides by 10

$$x = 4$$

Step 3: Check the answer by substituting the solution back into the original equation

$$\begin{aligned}\text{LHS} &= 4[2(4) - 9] - 4(4) \\ &= 4(8 - 9) - 16 \\ &= 4(-1) - 16 \\ &= -4 - 16 \\ &= -20\end{aligned}$$

$$\begin{aligned}\text{RHS} &= 4 - 6(4) \\ &= 4 - 24 \\ &= -20\end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$. Since both sides are equal, the answer is correct.

Worked example 2: Solving linear equations

QUESTION

Solve for x :

$$\frac{2-x}{3x+1} = 2$$

SOLUTION

Step 1: Multiply both sides of the equation by $(3x+1)$

Division by 0 is undefined so there must be a restriction: $(x \neq -\frac{1}{3})$.

$$\begin{aligned}\frac{2-x}{3x+1} &= 2 \\ (2-x) &= 2(3x+1)\end{aligned}$$

Step 2: Expand the brackets and simplify

$$\begin{aligned}2-x &= 6x+2 \\ -x-6x &= 2-2 \\ -7x &= 0\end{aligned}$$

Step 3: Divide both sides by -7

$$\begin{aligned}x &= \frac{0}{-7} \\ x &= 0\end{aligned}$$

Step 4: Check the answer by substituting the solution back into the original equation

$$\begin{aligned}\text{LHS} &= \frac{2-(0)}{3(0)+1} \\ &= 2 \\ &= \text{RHS}\end{aligned}$$

Since both sides are equal, the answer is correct.

QUESTION

Solve for a :

$$\frac{2a-3}{3} - 3a = \frac{a}{3}$$

SOLUTION

Step 1: Multiply the equation by the common denominator 3 and simplify

$$\begin{aligned} 2a - 3 - 9a &= a \\ -7a - 3 &= a \end{aligned}$$

Step 2: Rearrange the terms and simplify

$$\begin{aligned} -7a - a &= 3 \\ -8a &= 3 \end{aligned}$$

Step 3: Divide both sides by -8

$$a = -\frac{3}{8}$$

Step 4: Check the answer by substituting the solution back into the original equation

$$\begin{aligned} \text{LHS} &= \frac{2\left(-\frac{3}{8}\right) - 3}{3} - 3\left(-\frac{3}{8}\right) \\ &= \frac{\left(-\frac{3}{4}\right) - \frac{12}{4}}{3} + \frac{9}{8} \\ &= \left[-\frac{15}{4} \times \frac{1}{3}\right] + \frac{9}{8} \\ &= -\frac{5}{4} + \frac{9}{8} \\ &= -\frac{10}{8} + \frac{9}{8} \\ &= -\frac{1}{8} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{-\frac{3}{8}}{3} \\ &= \frac{-\frac{3}{8}}{3} \\ &= -\frac{3}{8} \times \frac{1}{3} \\ &= -\frac{1}{8} \end{aligned}$$

$\therefore$ LHS = RHS. Since both sides are equal, the answer is correct.

Exercise 4 – 1:

Solve the following equations (assume all denominators are non-zero):

1. $2y - 3 = 7$
2. $2c = c - 8$
3. $3 = 1 - 2c$
4. $4b + 5 = -7$
5. $-3y = 0$
6. $16y + 4 = -10$
7. $12y + 0 = 144$
8. $7 + 5y = 62$
9. $55 = 5x + \frac{3}{4}$
10. $5x = 2x + 45$
11. $23x - 12 = 6 + 3x$
12. $12 - 6x + 34x = 2x - 24 - 64$
13. $6x + 3x = 4 - 5(2x - 3)$
14. $18 - 2p = p + 9$
15. $\frac{4}{p} = \frac{16}{24}$
16. $-(-16 - p) = 13p - 1$
17. $3f - 10 = 10$
18. $3f + 16 = 4f - 10$
19. $10f + 5 = -2f - 3f + 80$
20. $8(f - 4) = 5(f - 4)$
21. $6 = 6(f + 7) + 5f$
22. $-7x = 8(1 - x)$
23. $5 - \frac{7}{b} = \frac{2(b + 4)}{b}$
24. $\frac{x + 2}{4} - \frac{x - 6}{3} = \frac{1}{2}$
25. $1 = \frac{3a - 4}{2a + 6}$
26. $\frac{2 - 5a}{3} - 6 = \frac{4a}{3} + 2 - a$
27. $2 - \frac{4}{b + 5} = \frac{3b}{b + 5}$
28. $3 - \frac{y - 2}{4} = 4$
29. $1,5x + 3,125 = 1,25x$
30. $1,3(2,7x + 1) = 4,1 - x$
31. $6,5x - 4,15 = 7 + 4,25x$
32. $\frac{1}{3}P + \frac{1}{2}P - 10 = 0$
33. $1\frac{1}{4}(x - 1) - 1\frac{1}{2}(3x + 2) = 0$
34. $\frac{1}{5}(x - 1) = \frac{1}{3}(x - 2) + 3$
35. $\frac{5}{2a} + \frac{1}{6a} - \frac{3}{a} = 2$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2F9C | 2. 2F9D | 3. 2F9F | 4. 2F9G | 5. 2F9H | 6. 2F9J | 7. 2F9K |
| 8. 2F9M | 9. 2F9N | 10. 2F9P | 11. 2F9Q | 12. 2F9R | 13. 2F9S | 14. 2F9T |
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| 29. 2FBC | 30. 2FBD | 31. 2FBF | 32. 2FBG | 33. 2FBH | 34. 2FBJ | 35. 2FBK |



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4.3 Solving quadratic equations

EMA36

A quadratic equation is an equation where the exponent of the variable is at most 2. The following are examples of quadratic equations:

$$\begin{aligned}2x^2 + 2x &= 1 \\3x^2 + 2x - 1 &= 0 \\0 &= -2x^2 + 4x - 2\end{aligned}$$

Quadratic equations differ from linear equations in that a linear equation has only one solution, while a quadratic equation has at most two solutions. There are some special situations, however, in which a quadratic equation has either one solution or no solutions.

We solve quadratic equations using factorisation. For example, in order to solve $2x^2 - x - 3 = 0$, we need to write it in its equivalent factorised form as $(x + 1)(2x - 3) = 0$. Note that if $a \times b = 0$ then $a = 0$ or $b = 0$.

VISIT:

The following video shows an example of solving a quadratic equation by factorisation.

► See video: [2FBM](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Method for solving quadratic equations

EMA37

1. Rewrite the equation in the required form, $ax^2 + bx + c = 0$.
2. Divide the entire equation by any common factor of the coefficients to obtain an equation of the form $ax^2 + bx + c = 0$, where a , b and c have no common factors. For example $2x^2 + 4x + 2 = 0$ can be written as $x^2 + 2x + 1 = 0$.
3. Factorise $ax^2 + bx + c = 0$ to be of the form $(rx + s)(ux + v) = 0$.
4. The two solutions are $(rx + s) = 0$ or $(ux + v) = 0$, so $x = -\frac{s}{r}$ or $x = -\frac{v}{u}$, respectively.
5. Check the answer by substituting it back into the original equation.

Worked example 4: Solving quadratic equations

QUESTION

Solve for x :

$$3x^2 + 2x - 1 = 0$$

SOLUTION

Step 1: The equation is already in the required form, $ax^2 + bx + c = 0$

Step 2: Factorise

$$(x + 1)(3x - 1) = 0$$

Step 3: Solve for both factors

We have:

$$x + 1 = 0$$

$$\therefore x = -1$$

OR

$$3x - 1 = 0$$

$$\therefore x = \frac{1}{3}$$

Step 4: Check both answers by substituting back into the original equation

Step 5: Write the final answer

The solution to $3x^2 + 2x - 1 = 0$ is $x = -1$ or $x = \frac{1}{3}$.

Worked example 5: Solving quadratic equations

QUESTION

Find the roots:

$$0 = -2x^2 + 4x - 2$$

SOLUTION

Step 1: The equation is already in the required form, $ax^2 + bx + c = 0$

Step 2: Divide the equation by common factor -2

$$-2x^2 + 4x - 2 = 0$$

$$x^2 - 2x + 1 = 0$$

Step 3: Factorise

$$(x - 1)(x - 1) = 0$$

$$(x - 1)^2 = 0$$

Step 4: The quadratic is a perfect square

This is an example of a special situation in which there is only one solution to the quadratic equation because both factors are the same.

$$x - 1 = 0$$

$$\therefore x = 1$$

Step 5: Check the answer by substituting back into the original equation

Step 6: Write final answer

The solution to $0 = -2x^2 + 4x - 2$ is $x = 1$.

Exercise 4 – 2:

1. Write the following in standard form:

a) $(r + 4)(5r - 4) = -16$

b) $(3r - 8)(2r - 3) = -15$

c) $(d + 5)(2d + 5) = 8$

2. Solve the following equations:

a) $x^2 + 2x - 15 = 0$

b) $p^2 - 7p - 18 = 0$

c) $9x^2 - 6x - 8 = 0$

d) $5x^2 + 21x - 54 = 0$

e) $4z^2 + 12z + 8 = 0$

f) $-b^2 + 7b - 12 = 0$

g) $-3a^2 + 27a - 54 = 0$

h) $4y^2 - 9 = 0$

i) $4x^2 + 16x - 9 = 0$
 k) $20m + 25m^2 = 0$
 m) $-75x^2 + 290x = 240$
 o) $x^2 - 4x = -4$
 q) $t^2 = 3t$
 s) $x^2 = 18$
 u) $4x^2 - 17x - 77 = 0$
 w) $2x^2 - 2x = 12$
 y) $(x - 6)^2 - 24 = 1$

j) $4x^2 - 12x = -9$
 l) $2x^2 - 5x - 12 = 0$
 n) $2x = \frac{1}{3}x^2 - 3x + 14\frac{2}{3}$
 p) $-x^2 + 4x - 6 = 4x^2 - 14x + 3$
 r) $x^2 - 10x = -25$
 t) $p^2 - 6p = 7$
 v) $14x^2 + 5x = 6$
 x) $(2a - 3)^2 - 16 = 0$

3. Solve the following equations (note the restrictions that apply):

a) $3y = \frac{54}{2y}$
 c) $x + 2 = \frac{18}{x} - 1$
 e) $\frac{1}{2}(b - 1) = \frac{1}{3}\left(\frac{2}{b} + 4\right)$
 g) $(x + 1)^2 - 2(x + 1) - 15 = 0$
 i) $b^4 - 13b^2 + 36 = 0$
 k) $\frac{x^2 - 2x - 3}{x + 1} = 0$
 m) $\frac{3(a^2 + 1) + 10a}{3a + 1} = 1$

b) $\frac{10z}{3} = 1 - \frac{1}{3z}$
 d) $y - 3 = \frac{5}{4} - \frac{1}{y}$
 f) $3(y + 1) = \frac{4}{y} + 2$
 h) $z^4 - 1 = 0$
 j) $\frac{a + 1}{3a - 4} + \frac{9}{2a + 5} + \frac{2a + 3}{2a + 5} = 0$
 l) $x + 2 = \frac{6x - 12}{x - 2}$
 n) $\frac{3}{9a^2 - 3a + 1} - \frac{3a + 4}{27a^3 + 1} = \frac{1}{9a^2 - 1}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2FBN	1b. 2FBP	1c. 2FBQ	2a. 2FBR	2b. 2FBS	2c. 2FBT	2d. 2FBV	2e. 2FBW
2f. 2FBX	2g. 2FBY	2h. 2FBZ	2i. 2FC2	2j. 2FC3	2k. 2FC4	2l. 2FC5	2m. 2FC6
2n. 2FC7	2o. 2FC8	2p. 2FC9	2q. 2FCB	2r. 2FCC	2s. 2FCD	2t. 2FCF	2u. 2FCG
2v. 2FCH	2w. 2FCJ	2x. 2FCK	2y. 2FCM	3a. 2FCN	3b. 2FCP	3c. 2FCQ	3d. 2FCR
3e. 2FCS	3f. 2FCT	3g. 2FCV	3h. 2FCW	3i. 2FCX	3j. 2FCY	3k. 2FCZ	3l. 2FD2
3m. 2FD3	3n. 2FD4						



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4.4 Solving simultaneous equations

EMA38

Up to now we have solved equations with only one unknown variable. When solving for two unknown variables, two equations are required and these equations are known as simultaneous equations. The solutions are the values of the unknown variables which satisfy both equations simultaneously. In general, if there are n unknown variables, then n independent equations are required to obtain a value for each of the n variables.

An example of a system of simultaneous equations is:

$$\begin{aligned}x + y &= -1 \\ 3 &= y - 2x\end{aligned}$$

We have two independent equations to solve for two unknown variables. We can solve simultaneous equations algebraically using substitution and elimination methods. We will also show that a system of simultaneous equations can be solved graphically.

- Use the simplest of the two given equations to express one of the variables in terms of the other.
- Substitute into the second equation. By doing this we reduce the number of equations and the number of variables by one.
- We now have one equation with one unknown variable which can be solved.
- Use the solution to substitute back into the first equation to find the value of the other unknown variable.

VISIT:

The following video shows how to solve simultaneous equations using substitution.

▶ See video: 2FD5 at www.everythingmaths.co.za

Worked example 6: Simultaneous equations**QUESTION**

Solve for x and y :

$$\begin{aligned}x - y &= 1 && \dots (1) \\3 &= y - 2x && \dots (2)\end{aligned}$$

SOLUTION

Step 1: Use equation (1) to express x in terms of y

$$x = y + 1$$

Step 2: Substitute x into equation (2) and solve for y

$$\begin{aligned}3 &= y - 2(y + 1) \\3 &= y - 2y - 2 \\5 &= -y \\\therefore y &= -5\end{aligned}$$

Step 3: Substitute y back into equation (1) and solve for x

$$\begin{aligned}x &= (-5) + 1 \\\therefore x &= -4\end{aligned}$$

Step 4: Check the solution by substituting the answers back into both original equations

Step 5: Write the final answer

$$\begin{aligned}x &= -4 \\y &= -5\end{aligned}$$

QUESTION

Solve the following system of equations:

$$\begin{aligned} 4y + 3x &= 100 & \dots (1) \\ 4y - 19x &= 12 & \dots (2) \end{aligned}$$

SOLUTION

Step 1: Use either equation to express x in terms of y

$$\begin{aligned} 4y + 3x &= 100 \\ 3x &= 100 - 4y \\ x &= \frac{100 - 4y}{3} \end{aligned}$$

Step 2: Substitute x into equation (2) and solve for y

$$\begin{aligned} 4y - 19 \left(\frac{100 - 4y}{3} \right) &= 12 \\ 12y - 19(100 - 4y) &= 36 \\ 12y - 1900 + 76y &= 36 \\ 88y &= 1936 \\ \therefore y &= 22 \end{aligned}$$

Step 3: Substitute y back into equation (1) and solve for x

$$\begin{aligned} x &= \frac{100 - 4(22)}{3} \\ &= \frac{100 - 88}{3} \\ &= \frac{12}{3} \\ \therefore x &= 4 \end{aligned}$$

Step 4: Check the solution by substituting the answers back into both original equations

Step 5: Write the final answer

$$\begin{aligned} x &= 4 \\ y &= 22 \end{aligned}$$

Worked example 8: Simultaneous equations**QUESTION**

Solve the following system of equations:

$$3x + y = 2 \quad \dots (1)$$

$$6x - y = 25 \quad \dots (2)$$

SOLUTION**Step 1: Make the coefficients of one of the variables the same in both equations**

The coefficients of y in the given equations are 1 and -1 . Eliminate the variable y by adding equation (1) and equation (2) together:

$$\begin{array}{rcl} 3x + y & = & 2 \\ + \quad 6x - y & = & 25 \\ \hline 9x + 0 & = & 27 \end{array}$$

Step 2: Simplify and solve for x

$$9x = 27$$

$$\therefore x = 3$$

Step 3: Substitute x back into either original equation and solve for y

$$3(3) + y = 2$$

$$y = 2 - 9$$

$$\therefore y = -7$$

Step 4: Check that the solution $x = 3$ and $y = -7$ satisfies both original equations**Step 5: Write the final answer**

$$x = 3$$

$$y = -7$$

QUESTION

Solve the following system of equations:

$$\begin{aligned} 2a - 3b &= 5 & \dots (1) \\ 3a - 2b &= 20 & \dots (2) \end{aligned}$$

SOLUTION

Step 1: Make the coefficients of one of the variables the same in both equations

By multiplying equation (1) by 3 and equation (2) by 2, both coefficients of a will be 6.

$$\begin{array}{rcl} 6a - 9b & = & 15 \\ - (6a - 4b & = & 40) \\ \hline 0 - 5b & = & -25 \end{array}$$

(When subtracting two equations, be careful of the signs.)

Step 2: Simplify and solve for b

$$\begin{aligned} b &= \frac{-25}{-5} \\ \therefore b &= 5 \end{aligned}$$

Step 3: Substitute value of b back into either original equation and solve for a

$$\begin{aligned} 2a - 3(5) &= 5 \\ 2a - 15 &= 5 \\ 2a &= 20 \\ \therefore a &= 10 \end{aligned}$$

Step 4: Check that the solution $a = 10$ and $b = 5$ satisfies both original equations

Step 5: Write the final answer

$$\begin{aligned} a &= 10 \\ b &= 5 \end{aligned}$$

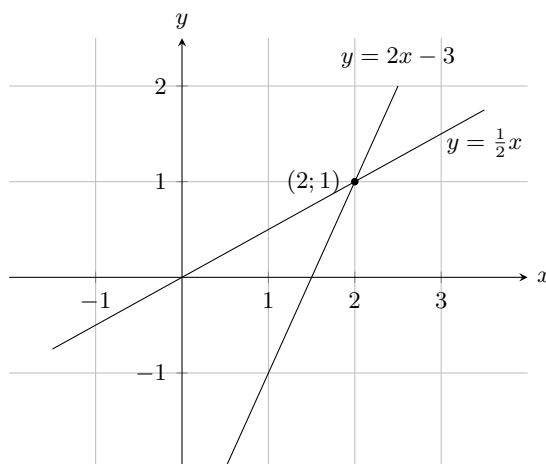
Simultaneous equations can also be solved graphically. If the graphs of each linear equation are drawn, then the solution to the system of simultaneous equations is the coordinates of the point at which the two graphs intersect.

For example:

$$x = 2y \quad \dots (1)$$

$$y = 2x - 3 \quad \dots (2)$$

The graphs of the two equations are shown below.



The intersection of the two graphs is (2; 1). So the solution to the system of simultaneous equations is $x = 2$ and $y = 1$. We can also check the solution using algebraic methods.

Substitute equation (1) into (2):

$$\begin{aligned} x &= 2y \\ \therefore y &= 2(2y) - 3 \end{aligned}$$

Then solve for y :

$$\begin{aligned} y - 4y &= -3 \\ -3y &= -3 \\ \therefore y &= 1 \end{aligned}$$

Substitute the value of y back into equation (1):

$$\begin{aligned} x &= 2(1) \\ \therefore x &= 2 \end{aligned}$$

Notice that both methods give the same solution.

VISIT:

You can use an online tool such as [graphsketch](#) to draw the graphs and check your solution.

QUESTION

Solve the following system of simultaneous equations graphically:

$$\begin{aligned} 4y + 3x &= 100 & \dots (1) \\ 4y - 19x &= 12 & \dots (2) \end{aligned}$$

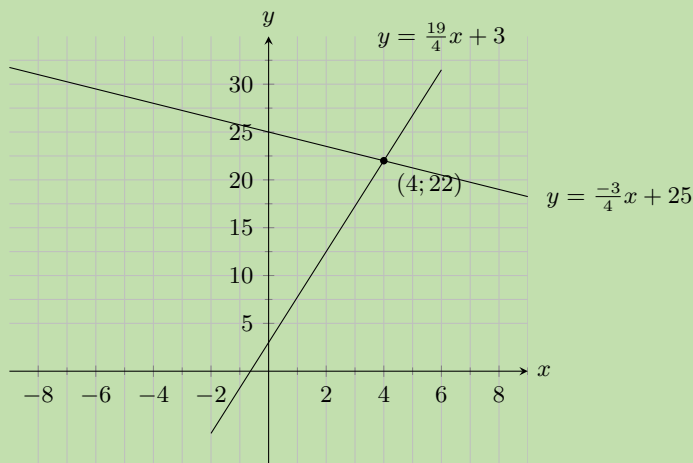
SOLUTION

Step 1: Write both equations in form $y = mx + c$

$$\begin{aligned} 4y + 3x &= 100 \\ 4y &= 100 - 3x \\ y &= -\frac{3}{4}x + 25 \end{aligned}$$

$$\begin{aligned} 4y - 19x &= 12 \\ 4y &= 19x + 12 \\ y &= \frac{19}{4}x + 3 \end{aligned}$$

Step 2: Sketch the graphs on the same set of axes



Step 3: Find the coordinates of the point of intersection

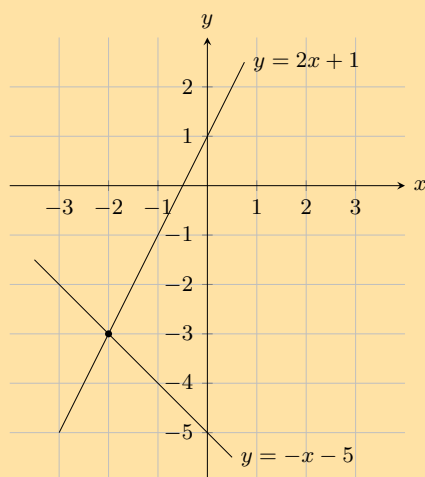
The two graphs intersect at (4; 22)

Step 4: Write the final answer

$$\begin{aligned} x &= 4 \\ y &= 22 \end{aligned}$$

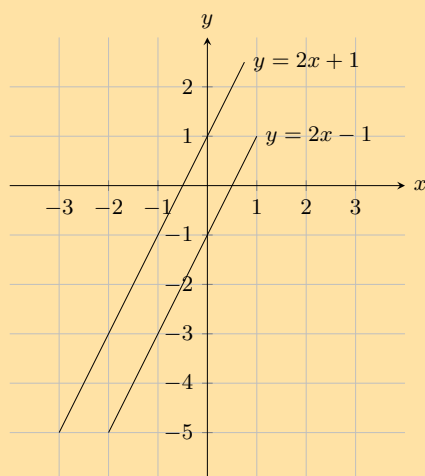
Exercise 4 – 3:

1. Look at the graph below:



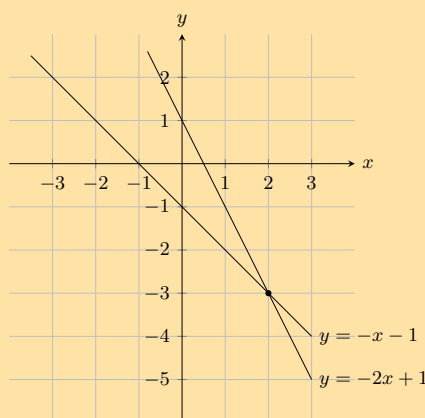
Solve the equations $y = 2x + 1$ and $y = -x - 5$ simultaneously.

2. Look at the graph below:



Solve the equations $y = 2x - 1$ and $y = 2x + 1$ simultaneously.

3. Look at the graph below:



Solve the equations $y = -2x + 1$ and $y = -x - 1$ simultaneously.

4. Solve for x and y :

a) $-10x = -1$ and $-4x + 10y = -9$.

c) $x + y = 8$ and $3x + 2y = 21$

e) $5x - 4y = 69$ and $2x + 3y = 23$

g) $3x - 4y = 19$ and $2x - 8y = 2$

i) $-10x + y = -1$ and $-10x - 2y = 5$

k) $\frac{1}{x} + \frac{1}{y} = 3$ and $\frac{1}{x} - \frac{1}{y} = 11$

m) $3a + b = \frac{6}{2a}$ and $3a^2 = 3 - ab$

b) $3x - 14y = 0$ and $x - 4y + 1 = 0$

d) $y = 2x + 1$ and $x + 2y + 3 = 0$

f) $x + 3y = 26$ and $5x + 4y = 75$

h) $\frac{a}{2} + b = 4$ and $\frac{a}{4} - \frac{b}{4} = 1$

j) $-10x - 10y = -2$ and $2x + 3y = 2$

l) $y = \frac{2(x^2 + 2) - 3}{x^2 + 2}$ and $y = 2 - \frac{3}{x^2 + 2}$

5. Solve graphically and check your answer algebraically:

a) $y + 2x = 0$ and $y - 2x - 4 = 0$

b) $x + 2y = 1$ and $\frac{x}{3} + \frac{y}{2} = 1$

c) $y - 2 = 6x$ and $y - x = -3$

d) $2x + y = 5$ and $3x - 2y = 4$

e) $5 = x + y$ and $x = y - 2$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. [2FD6](#)

2. [2FD7](#)

3. [2FD8](#)

4a. [2FDM](#)

4b. [2FD9](#)

4c. [2FDB](#)

4d. [2FDC](#)

4e. [2FDD](#)

4f. [2FDF](#)

4g. [2FDG](#)

4h. [2FDH](#)

4i. [2FDJ](#)

4j. [2FDK](#)

4k. [2FDN](#)

4l. [2FDP](#)

4m. [2FDQ](#)

5a. [2FDT](#)

5b. [2FDR](#)

5c. [2FDV](#)

5d. [2FDW](#)

5e. [2FDS](#)



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4.5 Word problems

EMA3D

To solve word problems we need to write a set of equations that represent the problem mathematically. The solution of the equations is then the solution to the problem.

Problem solving strategy

EMA3F

1. Read the whole question.
2. What are we asked to solve for?
3. Assign a variable to the unknown quantity, for example, x .
4. Translate the words into algebraic expressions by rewriting the given information in terms of the variable.
5. Set up an equation or system of equations to solve for the variable.
6. Solve the equation algebraically using substitution.
7. Check the solution.

VISIT:

The following video shows two examples of working with word problems.

▶ See video: [2FDX](#) at www.everythingmaths.co.za

Worked example 11: Solving word problems

QUESTION

A shop sells bicycles and tricycles. In total there are 7 cycles (cycles include both bicycles and tricycles) and 19 wheels. Determine how many of each there are, if a bicycle has two wheels and a tricycle has three wheels.

SOLUTION

Step 1: Assign variables to the unknown quantities

Let b be the number of bicycles and let t be the number of tricycles.

Step 2: Set up the equations

$$\begin{aligned}b + t &= 7 & \dots (1) \\2b + 3t &= 19 & \dots (2)\end{aligned}$$

Step 3: Rearrange equation (1) and substitute into equation (2)

$$\begin{aligned}t &= 7 - b \\ \therefore 2b + 21 - 3b &= 19 \\ -b &= -2 \\ \therefore b &= 2\end{aligned}$$

Step 4: Calculate the number of tricycles t

$$\begin{aligned}t &= 7 - b \\ &= 7 - 2 \\ &= 5\end{aligned}$$

Step 5: Write the final answer

There are 5 tricycles and 2 bicycles.

Worked example 12: Solving word problems

QUESTION

Bongani and Jane are friends. Bongani takes Jane's maths test paper and will not tell her what her mark is. He knows that Jane dislikes word problems so he decides to tease her. Bongani says: "I have 2 marks more than you do and the sum of both our marks is equal to 14. What are our marks?"

SOLUTION

Step 1: Assign variables to the unknown quantities

We have two unknown quantities, Bongani's mark and Jane's mark. Let Bongani's mark be b and Jane's mark be j .

Step 2: Set up a system of equations

Bongani has 2 more marks than Jane.

$$b = j + 2 \quad \dots (1)$$

Both marks add up to 14.

$$b + j = 14 \quad \dots (2)$$

Step 3: Use equation (1) to express b in terms of j

$$b = j + 2$$

Step 4: Substitute into equation (2)

$$\begin{aligned}b + j &= 14 \\(j + 2) + j &= 14\end{aligned}$$

Step 5: Rearrange and solve for j

$$\begin{aligned}2j &= 14 - 2 \\&= 12 \\\therefore j &= 6\end{aligned}$$

Step 6: Substitute the value for j back into equation (1) and solve for b

$$\begin{aligned}b &= j + 2 \\&= 6 + 2 \\&= 8\end{aligned}$$

Step 7: Check that the solution satisfies both original equations

Step 8: Write the final answer

Bongani got 8 for his test and Jane got 6.

Worked example 13: Solving word problems

QUESTION

A fruitshake costs R 2,00 more than a chocolate milkshake. If 3 fruitshakes and 5 chocolate milkshakes cost R 78,00, determine the individual prices.

SOLUTION

Step 1: Assign variables to the unknown quantities

Let the price of a chocolate milkshake be x and let the price of a fruitshake be y .

Step 2: Set up a system of equations

$$\begin{aligned}y &= x + 2 && \dots (1) \\3y + 5x &= 78 && \dots (2)\end{aligned}$$

Step 3: Substitute equation (1) into (2)

$$3(x + 2) + 5x = 78$$

Step 4: Rearrange and solve for x

$$3x + 6 + 5x = 78$$

$$8x = 72$$

$$\therefore x = 9$$

Step 5: Substitute the value of x back into equation (1) and solve for y

$$y = x + 2$$

$$= 9 + 2$$

$$= 11$$

Step 6: Check that the solution satisfies both original equations

Step 7: Write final answer

One chocolate milkshake costs R 9,00 and one fruitshake costs R 11,00.

Worked example 14: Solving word problems

QUESTION

The product of two consecutive negative integers is 1122. Find the two integers.

SOLUTION

Step 1: Assign variables to the unknown quantities

Let the first integer be n and let the second integer be $n + 1$

Step 2: Set up an equation

$$n(n + 1) = 1122$$

Step 3: Expand and solve for n

$$n^2 + n = 1122$$

$$n^2 + n - 1122 = 0$$

$$(n + 34)(n - 33) = 0$$

$$\therefore n = -34$$

$$\text{or } n = 33$$

Step 4: Find the sign of the integers

It is given that both integers must be negative.

$$\begin{aligned}\therefore n &= -34 \\ n + 1 &= -34 + 1 \\ &= -33\end{aligned}$$

Step 5: Write the final answer

The two consecutive negative integers are -34 and -33 .

Exercise 4 – 4:

- Two jets are flying towards each other from airports that are 1200 km apart. One jet is flying at $250 \text{ km}\cdot\text{h}^{-1}$ and the other jet at $350 \text{ km}\cdot\text{h}^{-1}$. If they took off at the same time, how long will it take for the jets to pass each other?
- Two boats are moving towards each other from harbours that are 144 km apart. One boat is moving at $63 \text{ km}\cdot\text{h}^{-1}$ and the other boat at $81 \text{ km}\cdot\text{h}^{-1}$. If both boats started their journey at the same time, how long will they take to pass each other?
- Zwelibanzi and Jessica are friends. Zwelibanzi takes Jessica's civil technology test paper and will not tell her what her mark is. He knows that Jessica dislikes word problems so he decides to tease her. Zwelibanzi says: "I have 12 marks more than you do and the sum of both our marks is equal to 148. What are our marks?"
- Kadesh bought 20 shirts at a total cost of R 980. If the large shirts cost R 50 and the small shirts cost R 40, how many of each size did he buy?
- The diagonal of a rectangle is 25 cm more than its width. The length of the rectangle is 17 cm more than its width. What are the dimensions of the rectangle?
- The sum of 27 and 12 is equal to 73 more than an unknown number. Find the unknown number.
- A group of friends is buying lunch. Here are some facts about their lunch:
 - a milkshake costs R 7 more than a wrap
 - the group buys 8 milkshakes and 2 wraps
 - the total cost for the lunch is R 326

Determine the individual prices for the lunch items.

- The two smaller angles in a right-angled triangle are in the ratio of 1 : 2. What are the sizes of the two angles?
- The length of a rectangle is twice the breadth. If the area is 128 cm^2 , determine the length and the breadth.
- If 4 times a number is increased by 6, the result is 15 less than the square of the number. Find the number.
- The length of a rectangle is 2 cm more than the width of the rectangle. The perimeter of the rectangle is 20 cm. Find the length and the width of the rectangle.
- Stephen has 1 litre of a mixture containing 69% salt. How much water must Stephen add to make the mixture 50% salt? Write your answer as a fraction of a litre.
- The sum of two consecutive odd numbers is 20 and their difference is 2. Find the two numbers.
- The denominator of a fraction is 1 more than the numerator. The sum of the fraction and its reciprocal is $\frac{5}{2}$. Find the fraction.

15. Masindi is 21 years older than her daughter, Mulivhu. The sum of their ages is 37. How old is Mulivhu?
16. Tshamano is now five times as old as his son Murunwa. Seven years from now, Tshamano will be three times as old as his son. Find their ages now.
17. If adding one to three times a number is the same as the number, what is the number equal to?
18. If a third of the sum of a number and one is equivalent to a fraction whose denominator is the number and numerator is two, what is the number?
19. A shop owner buys 40 sacks of rice and mealie meal worth R 5250 in total. If the rice costs R 150 per sack and mealie meal costs R 100 per sack, how many sacks of mealie meal did he buy?
20. There are 100 bars of blue and green soap in a box. The blue bars weigh 50 g per bar and the green bars 40 g per bar. The total mass of the soap in the box is 4,66 kg. How many bars of green soap are in the box?
21. Lisa has 170 beads. She has blue, red and purple beads each weighing 13 g, 4 g and 8 g respectively. If there are twice as many red beads as there are blue beads and all the beads weigh 1,216 kg, how many beads of each type does Lisa have?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2FDY | 2. 2FDZ | 3. 2FF2 | 4. 2FF3 | 5. 2FF4 | 6. 2FF5 | 7. 2FF6 | 8. 2FF7 |
| 9. 2FF8 | 10. 2FF9 | 11. 2FFB | 12. 2FFC | 13. 2FFD | 14. 2FFF | 15. 2FFG | 16. 2FFH |
| 17. 2FFJ | 18. 2FFK | 19. 2FFM | 20. 2FFN | 21. 2FFP | | | |



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4.6 Literal equations

EMA3G

A literal equation is one that has several letters or variables. Examples include the area of a circle ($A = \pi r^2$) and the formula for speed ($v = \frac{D}{t}$). In this section we solve literal equations in terms of one variable. To do this, we use the principles we have learnt about solving equations and apply them to rearranging literal equations. Solving literal equations is also known as changing the subject of the formula.

Keep the following in mind when solving literal equations:

- We isolate the unknown variable by asking “what is it joined to?” and “how is it joined?” We then perform the opposite operation to both sides as a whole.
- If the unknown variable is in two or more terms, then we take it out as a common factor.
- If we have to take the square root of both sides, remember that there will be a positive and a negative answer.
- If the unknown variable is in the denominator, we multiply both sides by the lowest common denominator (LCD) and then continue to solve.

VISIT:

The following video shows an example of solving literal equations.

▶ See video: [2FFQ](#) at www.everythingmaths.co.za

Worked example 15: Solving literal equations

QUESTION

The area of a triangle is $A = \frac{1}{2}bh$. What is the height of the triangle in terms of the base and area?

SOLUTION

Step 1: Isolate the required variable

We are asked to isolate the height, so we must rearrange the equation with h on one side of the equals sign and the rest of the variables on the other.

$$\begin{aligned}A &= \frac{1}{2}bh \\2A &= bh \\ \frac{2A}{b} &= h\end{aligned}$$

Step 2: Write the final answer

The height of a triangle is given by: $h = \frac{2A}{b}$

Worked example 16: Solving literal equations

QUESTION

Given the formula:

$$h = R \times \frac{H}{R + r^2}$$

make R the subject of the formula.

SOLUTION

Step 1: Isolate the required variable

$$\begin{aligned}h(R + r^2) &= R \times H \\hR + hr^2 &= HR \\hr^2 &= HR - hR \\hr^2 &= R(H - h) \\\therefore R &= \frac{hr^2}{H - h}\end{aligned}$$

Exercise 4 – 5:

1. Solve for x in the following formula: $2x + 4y = 2$.
2. Make a the subject of the formula: $s = ut + \frac{1}{2}at^2$.
3. Solve for n : $pV = nRT$.
4. Make x the subject of the formula: $\frac{1}{b} + \frac{2b}{x} = 2$.
5. Solve for r : $V = \pi r^2 h$.
6. Solve for h : $E = \frac{hc}{\lambda}$.

7. Solve for h : $A = 2\pi rh + 2\pi r$.
8. Make λ the subject of the formula: $t = \frac{D}{f\lambda}$.
9. Solve for m : $E = mgh + \frac{1}{2}mv^2$.
10. Solve for x : $x^2 + x(a + b) + ab = 0$.
11. Solve for b : $c = \sqrt{a^2 + b^2}$.
12. Make U the subject of the formula: $\frac{1}{V} = \frac{1}{U} + \frac{1}{W}$.
13. Solve for r : $A = \pi R^2 - \pi r^2$.
14. $F = \frac{9}{5}C + 32^\circ$ is the formula for converting temperature in degrees Celsius to degrees Fahrenheit. Derive a formula for converting degrees Fahrenheit to degrees Celsius.
15. $V = \frac{4}{3}\pi r^3$ is the formula for determining the volume of a soccer ball. Express the radius in terms of the volume.
16. Solve for x in: $x^2 - ax - 3x = 4 + a$
17. Solve for x in: $ax^2 - 4a + bx^2 - 4b = 0$
18. Solve for x in $v^2 = u^2 + 2ax$ if $v = 2$, $u = 0,3$, $a = 0,5$
19. Solve for u in $f' = f \frac{v}{v - u}$ if $v = 13$, $f = 40$, $f' = 50$
20. Solve for h in $I = \frac{bh^2}{12}$ if $b = 18$, $I = 384$
21. Solve for r_2 in $\frac{1}{R} = \frac{1}{r_1} + \frac{1}{r_2}$ if $R = \frac{3}{2}$, $r_1 = 2$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2FFR | 2. 2FFS | 3. 2FFT | 4. 2FFV | 5. 2FFW | 6. 2FFX | 7. 2FFY | 8. 2FFZ |
| 9. 2FG2 | 10. 2FG3 | 11. 2FG4 | 12. 2FG5 | 13. 2FG6 | 14. 2FG7 | 15. 2FG8 | 16. 2FG9 |
| 17. 2FGB | 18. 2FGC | 19. 2FGD | 20. 2FGF | 21. 2FGG | | | |



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4.7 Solving linear inequalities

EMA3H

A linear inequality is similar to a linear equation in that the largest exponent of a variable is 1. The following are examples of linear inequalities.

$$2x + 2 \leq 1$$

$$\frac{2 - x}{3x + 1} \geq 2$$

$$\frac{4}{3}x - 6 < 7x + 2$$

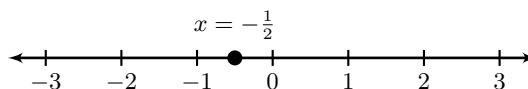
The methods used to solve linear inequalities are similar to those used to solve linear equations. The only difference occurs when there is a multiplication or a division that involves a minus sign. For example, we know that $8 > 6$. If both sides of the inequality are divided by -2 , then we get $-4 > -3$, which is not true. Therefore, the inequality sign must be switched around, giving $-4 < -3$.

In order to compare an inequality to a normal equation, we shall solve an equation first.

Solve $2x + 2 = 1$:

$$\begin{aligned} 2x + 2 &= 1 \\ 2x &= 1 - 2 \\ 2x &= -1 \\ x &= -\frac{1}{2} \end{aligned}$$

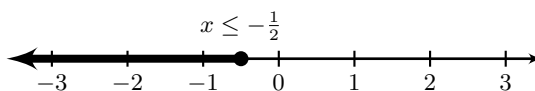
If we represent this answer on a number line, we get:



Now let us solve for x in the inequality $2x + 2 \leq 1$:

$$\begin{aligned} 2x + 2 &\leq 1 \\ 2x &\leq 1 - 2 \\ 2x &\leq -1 \\ x &\leq -\frac{1}{2} \end{aligned}$$

If we represent this answer on a number line, we get:



We see that for the equation there is only a single value of x for which the equation is true. However, for the inequality, there is a range of values for which the inequality is true. This is the main difference between an equation and an inequality.

Remember: when we divide or multiply both sides of an inequality by a negative number, the direction of the inequality changes. For example, if $x < 1$, then $-x > -1$. Also note that we cannot divide or multiply by a variable.

NOTE:

The following video provides an introduction to linear inequalities.

► See video: [2FGH](#) at www.everythingmaths.co.za

Interval notation

EMA3J

Examples:

$(4; 12)$	Round brackets indicate that the number is not included. This interval includes all real numbers greater than but not equal to 4 and less than but not equal to 12.
$(-\infty; -1)$	Round brackets are always used for positive and negative infinity. This interval includes all real numbers less than, but not equal to -1 .
$[1; 13)$	A square bracket indicates that the number is included. This interval includes all real numbers greater than or equal to 1 and less than but not equal to 13.

It is important to note that this notation can only be used to represent an interval of real numbers.

We represent the above answer in interval notation as $(-\infty; -\frac{1}{2}]$

Worked example 17: Solving linear inequalities

QUESTION

Solve for r :

$$6 - r > 2$$

Represent the answer on a number line and in interval notation.

SOLUTION

Step 1: Rearrange and solve for r

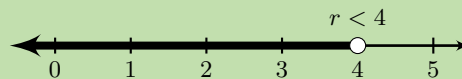
$$-r > 2 - 6$$

$$-r > -4$$

Step 2: Multiply by -1 and reverse inequality sign

$$r < 4$$

Step 3: Represent the answer on a number line



Step 4: Represent the answer in interval notation

$$(-\infty; 4)$$

Worked example 18: Solving linear inequalities

QUESTION

Solve for q :

$$4q + 3 < 2(q + 3)$$

Represent the answer on a number line and in interval notation.

SOLUTION

Step 1: Expand the bracket

$$4q + 3 < 2(q + 3)$$

$$4q + 3 < 2q + 6$$

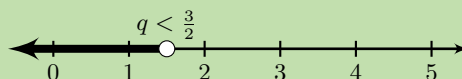
Step 2: Rearrange and solve for q

$$\begin{aligned}4q + 3 &< 2q + 6 \\4q - 2q &< 6 - 3 \\2q &< 3\end{aligned}$$

Step 3: Divide both sides by 2

$$\begin{aligned}2q &< 3 \\q &< \frac{3}{2}\end{aligned}$$

Step 4: Represent the answer on a number line



Step 5: Represent the answer in interval notation

$$\left(-\infty; \frac{3}{2}\right)$$

Worked example 19: Solving compound linear inequalities

QUESTION

Solve for x :

$$5 \leq x + 3 < 8$$

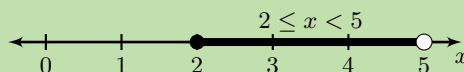
Represent the answer on a number line and in interval notation.

SOLUTION

Step 1: Subtract 3 from all the parts of the inequality

$$\begin{array}{rcccl}5 - 3 & \leq & x + 3 - 3 & < & 8 - 3 \\2 & \leq & x & < & 5\end{array}$$

Step 2: Represent the answer on a number line



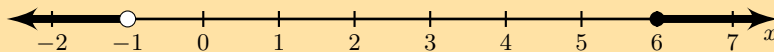
Step 3: Represent the answer in interval notation

$$[2; 5)$$

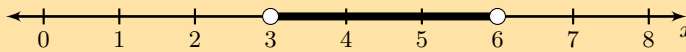
Exercise 4 – 6:

1. Look at the number line and write down the inequality it represents.

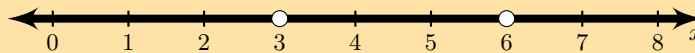
a)



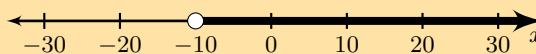
b)



c)



d)



2. Solve for x and represent the answer on a number line and in interval notation.

a) $3x + 4 > 5x + 8$

b) $3(x - 1) - 2 \leq 6x + 4$

c) $\frac{x - 7}{3} > \frac{2x - 3}{2}$

d) $-4(x - 1) < x + 2$

e) $\frac{1}{2}x + \frac{1}{3}(x - 1) \geq \frac{5}{6}x - \frac{1}{3}$

f) $-2 \leq x - 1 < 3$

g) $-5 < 2x - 3 \leq 7$

h) $7(3x + 2) - 5(2x - 3) > 7$

i) $\frac{5x - 1}{-6} \geq \frac{1 - 2x}{3}$

j) $3 \leq 4 - x \leq 16$

k) $\frac{-7y}{3} - 5 > -7$

l) $1 \leq 1 - 2y < 9$

m) $-2 < \frac{x - 1}{-3} < 7$

3. Solve for x and show your answer in interval notation:

a) $2x - 1 < 3(x + 11)$

b) $x - 1 < -4(x - 6)$

c) $\frac{x - 1}{8} \leq \frac{2(x - 2)}{3}$

d) $\frac{x + 2}{4} \leq \frac{-2(x - 4)}{7}$

e) $\frac{1}{5}x - \frac{5}{4}(x + 2) > \frac{1}{4}x + 3$

f) $\frac{1}{5}x - \frac{2}{5}(x + 3) \geq \frac{4}{2}x + 3$

g) $4x + 3 < -3$ or $4x + 3 > 5$

h) $4 \geq -6x - 6 \geq -3$

4. Solve for the unknown variable and show your answer on a number line.

a) $6b - 3 > b + 2, b \in \mathbb{Z}$

b) $3a - 1 < 4a + 6, a \in \mathbb{N}$

c) $\frac{b - 3}{2} + 1 < \frac{b}{4} - 4, b \in \mathbb{R}$

d) $\frac{4a + 7}{3} - 5 > a - \frac{2}{3}, a \in \mathbb{N}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. [2FGJ](#)

1b. [2FGK](#)

1c. [2FGM](#)

1d. [2FGN](#)

2a. [2FGP](#)

2b. [2FGQ](#)

2c. [2FGR](#)

2d. [2FGS](#)

2e. [2FGT](#)

2f. [2FGV](#)

2g. [2FGW](#)

2h. [2FGX](#)

2i. [2FGY](#)

2j. [2FGZ](#)

2k. [2FH2](#)

2l. [2FH3](#)

2m. [2FH4](#)

3a. [2FH5](#)

3b. [2FH6](#)

3c. [2FH7](#)

3d. [2FH8](#)

3e. [2FH9](#)

3f. [2FHB](#)

3g. [2FHC](#)

3h. [2FHD](#)

4a. [2FHF](#)

4b. [2FHG](#)

4c. [2FHH](#)

4d. [2FHJ](#)



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► See presentation: 2FHK at www.everythingmaths.co.za

- A linear equation is an equation where the exponent of the variable is 1. A linear equation has at most one solution.
- A quadratic equation is an equation where the exponent of the variable is at most 2. A quadratic equation has at most two solutions.
- To solve for two unknown variables, two equations are required. These equations are known as a system of simultaneous equations. There are two ways to solve linear simultaneous equations: algebraic solutions and graphical solutions. To solve algebraically we use substitution or elimination methods. To solve graphically we draw the graph of each equation and the solution will be the coordinates of the point of intersection.
- Literal equations are equations that have several letters and variables.
- Word problems require a set of equations that represent the problem mathematically.
- A linear inequality is similar to a linear equation and has the exponent of the variable equal to 1.
- If we divide or multiply both sides of an inequality by any number with a minus sign, the direction of the inequality changes.

End of chapter Exercise 4 – 7:

1. Solve:

a) $5a - 7 = -2$

c) $1 = 4 - 3y$

e) $3 - 6k = 2k - 1$

g) $m + 6(-m + 1) + 8m = 0$

i) $3 + \frac{q}{5} = \frac{q}{2}$

k) $2 + \frac{a-4}{2} - \frac{a}{3} = 7$

m) $\frac{2}{t} - 2 - \frac{1}{2} = \frac{1}{2} \left(1 + \frac{2}{t}\right)$

b) $5m + 3 = -2$

d) $2(p - 1) = 3(p + 2)$

f) $2,1x + 3 = 4,1 - 3,3x$

h) $2k + 3 = 2 - 3(k + 2)$

j) $\frac{1}{2} = \frac{4z + 1}{5z - 6}$

l) $5 - \frac{2(m + 4)}{m} = \frac{7}{m}$

2. Solve:

a) $b^2 + 6b - 27 = 0$

c) $-b^2 - 3b + 10 = 0$

e) $(5x + 1)(x - 3) = 0$

g) $\frac{a+2}{a-3} = \frac{a+8}{a+4}$

i) $x^2 - 3x + 2 = 0$

k) $0 = 2x^2 - 5x - 18$

b) $-x^2 + 5x + 6 = 0$

d) $2b - 15 = (b + 1)(b - 6) - b^2$

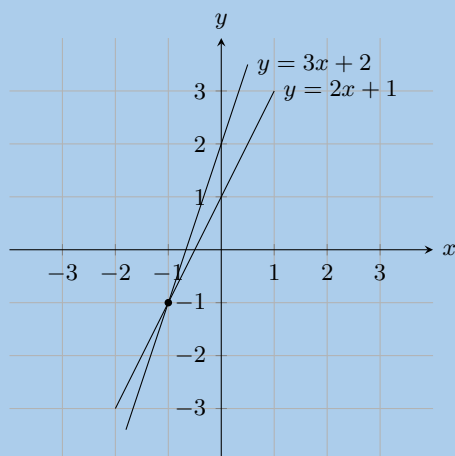
f) $5t - 1 = t^2 - (t + 2)(t - 2)$

h) $\frac{n+3}{n-2} = \frac{n-1}{n-7}$

j) $y^2 + y = 6$

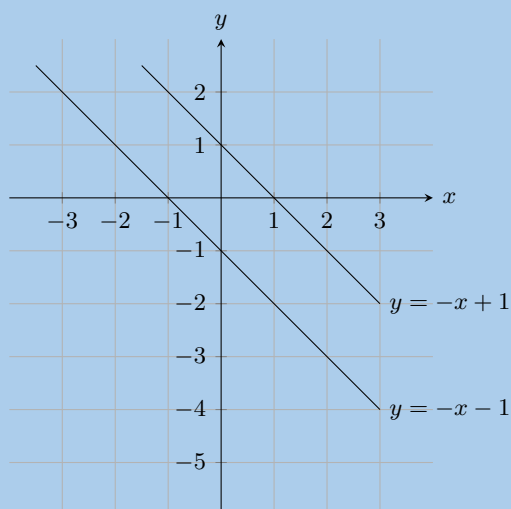
l) $(d + 4)(d - 3) - d = (3d - 2)^2 - 8d(d - 1)$

3. Look at the graph below:



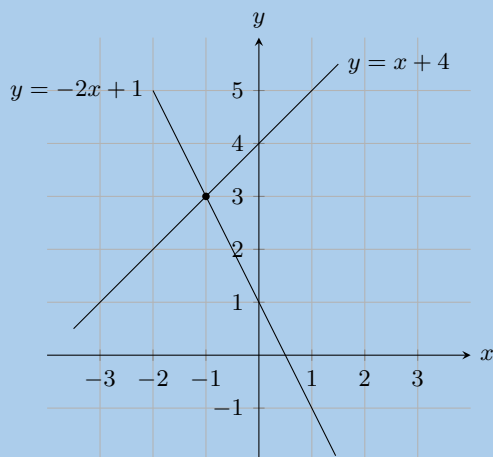
Solve the equations $y = 3x + 2$ and $y = 2x + 1$ simultaneously.

4. Look at the graph below:



Solve the equations $y = -x + 1$ and $y = -x - 1$ simultaneously.

5. Look at the graph below:



Solve the equations $y = x + 4$ and $y = -2x + 1$ simultaneously.

6. Solve the following simultaneous equations:

- a) $7x + 3y = 13$ and $2x - 3y = -4$ b) $10 = 2x + y$ and $y = x - 2$
c) $7x - 41 = 3y$ and $17 = 3x - y$ d) $2x - 4y = 32$ and $7x + 2y = 32$
e) $7x + 6y = -18$ and $4x + 12y = 24$ f) $3x - 4y = -15$ and $12x + 5y = 66$
g) $x - 3y = -22$ and $5x + 2y = -25$ h) $3x + 2y = 46$ and $15x + 5y = 220$
i) $6x + 3y = -63$ and $24x + 4y = -212$ j) $5x - 6y = 11$ and $25x - 3y = 28$
k) $-9x + 3y = 4$ and $2x + 2y = 6$ l) $3x - 7y = -10$ and $10x + 2y = -6$
m) $2y = x + 8$ and $4y = 2x - 44$ n) $2a(a - 1) - 4 + a - b = 0$ and $2a^2 - a = b + 4$
o) $y = (x - 2)^2$ and $x(x + 3) - y = 3x + 4(x - 1)$ p) $\frac{x + 1}{y} = 7$ and $\frac{x}{y + 1} = 6$
q) $(x + 3)^2 + (y - 4)^2 = 0$

7. Find the solutions to the following word problems:

- a) $\frac{7}{8}$ of a certain number is 5 more than $\frac{1}{3}$ of the number. Find the number.
b) Three rulers and two pens have a total cost of R 21,00. One ruler and one pen have a total cost of R 8,00. How much does a ruler cost and how much does a pen cost?
c) A group of friends is buying lunch. Here are some facts about their lunch:
- a hotdog costs R 6 more than a milkshake
 - the group buys 3 hotdogs and 2 milkshakes
 - the total cost for the lunch is R 143

Determine the individual prices for the lunch items.

- d) Lefu and Monique are friends. Monique takes Lefu's business studies test paper and will not tell him what his mark is. She knows that Lefu dislikes word problems so she decides to tease him. Monique says: "I have 12 marks more than you do and the sum of both our marks is equal to 166. What are our marks?"
e) A man runs to the bus stop and back in 15 minutes. His speed on the way to the bus stop is $5 \text{ km} \cdot \text{h}^{-1}$ and his speed on the way back is $4 \text{ km} \cdot \text{h}^{-1}$. Find the distance to the bus stop.
f) Two trucks are travelling towards each other from factories that are 175 km apart. One truck is travelling at $82 \text{ km} \cdot \text{h}^{-1}$ and the other truck at $93 \text{ km} \cdot \text{h}^{-1}$. If both trucks started their journey at the same time, how long will they take to pass each other?
g) Zanele and Piet skate towards each other on a straight path. They set off 20 km apart. Zanele skates at $15 \text{ km} \cdot \text{h}^{-1}$ and Piet at $10 \text{ km} \cdot \text{h}^{-1}$. How far will Piet have skated when they reach each other?
h) When the price of chocolates is increased by R 10, we can buy five fewer chocolates for R 300. What was the price of each chocolate before the price was increased?
i) A teacher bought R 11 300 worth of textbooks. The text books were for Science and Mathematics with each of them being sold at R 100 per book and R 125 per book respectively. If the teacher bought 97 books in total, how many Science books did she buy?
j) Thom's mom bought R 91,50 worth of easter eggs. The easter eggs came in 3 different colours blue, green and yellow. The blue ones cost R 2 each, green ones R 1,50 each and yellow ones R 1 each. She bought three times as many yellow eggs as the green ones and 72 eggs in total. How many blue eggs did she buy?
k) Two equivalent fractions both have their numerator as one. The denominator of one fraction is the sum of two and a number, while the other fraction is twice the number less 3. What is the number?

8. Consider the following literal equations:

- a) Solve for x : $a - bx = c$

b) Solve for I : $P = VI$.

c) Make m the subject of the formula: $E = mc^2$.

d) Solve for t : $v = u + at$.

e) Make f the subject of the formula: $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$.

f) Solve for y : $m = \frac{y - c}{x}$.

g) Solve for x in: $ax - 4a + ab = 4b - bx - b^2 + 4c - cx - bc$

h) Solve for r in $S = \frac{a}{1-r}$ if $a = 0,4$ and $S = 3$

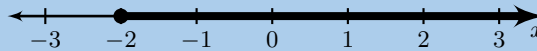
i) Solve for b in $I = \frac{1}{2}M(a^2 + b^2)$ if $a = 4$, $M = 8$, $I = 320$

9. Write down the inequality represented by the following:

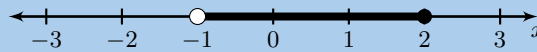
a)



b)



c)



10. Solve for x and show your answer in interval notation

a) $-4x + 1 > -2(x - 15)$

b) $\frac{x+2}{4} \leq \frac{-1(x+1)}{6}$

c) $\frac{1}{4}x + \frac{2}{3}(x+1) \geq \frac{2}{5}x + 2$

d) $3x - 3 > 14$ or $3x - 3 < -2$

11. Solve and represent your answer on a number line

a) $2x - 3 < \frac{3x - 2}{2}$, $x \in \mathbb{N}$

b) $3(1 - b) - 4 + b > 7 + b$, $b \in \mathbb{Z}$

c) $1 - 5x > 4(x + 1) - 3$, $x \in \mathbb{R}$

12. Solve for the unknown variable:

a) $2 + 2\frac{1}{3}(x + 4) = \frac{1}{5}(3 - x) + \frac{1}{6}$

b) $36 - (x - 4)^2 = 0$

c) $64 - (a + 3)^2 = 0$

d) $\frac{1}{2}x - \frac{2}{x} = 0$

e) $a - 3 = 2\left(\frac{6}{a} + 1\right)$

f) $a - \frac{6}{a} = -1$

g) $(a + 6)^2 - 5(a + 6) - 24 = 0$

h) $a^4 - 4a^2 + 3 = 0$

i) $9y^4 - 13y^2 + 4 = 0$

j) $\frac{(b+1)^2 - 16}{b+5} = 1$

k) $\frac{a^2 + 8a + 7}{a + 7} = 2$

l) $5x + 2 \leq 4(2x - 1)$

m) $\frac{4x - 2}{6} > 2x + 1$

n) $\frac{x}{3} - 14 > 14 - \frac{x}{7}$

$$o) \frac{1-a}{2} - \frac{2-a}{3} \geq 1$$

$$q) x - 1 = \frac{42}{x}$$

$$s) 3ax + 2a - ax = 5ax - 6a$$

$$u) 3x^2 - xy - 2y^2 = 0$$

$$w) \frac{2x-5}{(x+2)(x-4)} = \frac{1}{2(x-4)}$$

$$y) \frac{x+4}{3} - 2 > \frac{x-3}{2} - \frac{x+1}{4}$$

$$p) -5 \leq 2k + 1 < 5$$

$$r) (x+1)^2 = (x+1)(2x+3)$$

$$t) \frac{ax}{b} - \frac{bx}{a} = \frac{a}{b} + 1$$

$$v) x(2x+1) = 1$$

$$x) x^2 + 1 = 0$$

13. After solving an equation, Luke gave his answer as 4,5 rounded to one decimal digit. Show on a number line the interval in which his solution lay.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2FHM	1b. 2FHN	1c. 2FHP	1d. 2FHQ	1e. 2FHR	1f. 2FHS	1g. 2FHT
1h. 2FHV	1i. 2FHW	1j. 2FHX	1k. 2FHY	1l. 2FHZ	1m. 2FJ2	2a. 2FJ3
2b. 2FJ4	2c. 2FJ5	2d. 2FJ6	2e. 2FJ7	2f. 2FJ8	2g. 2FJ9	2h. 2FJB
2i. 2FJC	2j. 2FJD	2k. 2FJF	2l. 2FJG	3. 2FJH	4. 2FJJ	5. 2FJK
6a. 2FJM	6b. 2FJN	6c. 2FJP	6d. 2FJQ	6e. 2FJR	6f. 2FJS	6g. 2FJT
6h. 2FJV	6i. 2FJW	6j. 2FJX	6k. 2FJY	6l. 2FJZ	6m. 2FK2	6n. 2FK3
6o. 2FK4	6p. 2FK5	6q. 2FK6	7a. 2FK7	7b. 2FK8	7c. 2FK9	7d. 2FKB
7e. 2FKC	7f. 2FKD	7g. 2FKF	7h. 2FKG	7i. 2FKH	7j. 2FKJ	7k. 2FKK
8a. 2FKM	8b. 2FKN	8c. 2FKP	8d. 2FKQ	8e. 2FKR	8f. 2FKS	8g. 2FKT
8h. 2FKV	8i. 2FKW	9a. 2FKX	9b. 2FKY	9c. 2FKZ	10a. 2FM2	10b. 2FM3
10c. 2FM4	10d. 2FM5	11a. 2FM6	11b. 2FM7	11c. 2FM8	12a. 2FM9	12b. 2FMB
12c. 2FMC	12d. 2FMD	12e. 2FMF	12f. 2FMG	12g. 2FMH	12h. 2FMJ	12i. 2FMK
12j. 2FMM	12k. 2FMN	12l. 2FMP	12m. 2FMQ	12n. 2FMR	12o. 2FMS	12p. 2FMT
12q. 2FMV	12r. 2FMW	12s. 2FMX	12t. 2FMY	12u. 2FMZ	12v. 2FN2	12w. 2FN3
12x. 2FN4	12y. 2FN5	13. 2FN6				



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